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A Second Look: Local Labor Markets and The Impact of Ban the Box Policies After Criminal Legal Involvement

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Abstract

This paper estimates the impact of labor demand on the employment and recidivism outcomes of released prisoners. Higher labor demand at release generates higher earnings and lower recidivism. Reduced recidivism persists after controlling for the observed labor market outcomes of the returning cohort, suggesting that labor demand impacts crime through channels beyond the direct formal employment of returning prisoners. Difference-in-differences based evidence suggests Ban the Box (BTB) policies delaying when employers can ask about criminal records improve labor market outcomes and lower recidivism for misdemeanor defendants. Evidence for felony defendants and returning prisoners is mixed but suggestive of similar patterns.

Keywords: Economics of crime; Labor markets; Reentry; Ban the Box; Recidivism
JEL classification codes: J23; J78; K14; K31; K42; K40

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1 Introduction

Millions of United States citizens have criminal records. In 2017, 622,400 prisoners were released from state and federal prisons (Bronson and Carson, 2019). Many of these released people are of working age. Researchers estimate that 8% of all adults and 33% of the African American adult male population have a felony conviction (Shannon et al., 2017). There is substantial evidence that employers are less willing to hire applicants with criminal histories. However, there is more limited evidence about why employers are reluctant to hire workers with criminal records. Research has also demonstrated that employment challenges increase recidivism, hamper national productivity, exacerbate racial income inequality, increase crime rates, and cause a host of other problems.¹ Reduced earning potential will have inter-generational consequences. Finlay, Mueller-Smith, and Street (2022) document 9% of children with parental prison involvement, 18% with a parent with a felony conviction, and 39% with a criminal charge (62% for Black children), consequences of the criminal justice system reach a vast proportion of the population.

This paper estimates the impact of labor demand during the year of release on the employment and recidivism outcomes of people released from prisons across more than a dozen states and interrogates the mechanisms at play. Like previous research (Raphael and Winter-Ebmer, 2001; Yang, 2017; Schnepel, 2018), I find stronger labor markets at release decrease recidivism rates, but also present novel evidence that demand is reducing recidivism through channels beyond employment for returning citizens. Direct earnings measures allow us to learn about the mechanisms at play.

Several other policies may interact with the business cycle in determining the labor market prospects for job seekers with criminal records. This paper presents new evidence on one popular policy designed to give a second chance to people with records: Ban the Box (BTB). BTB may be an important confounder if it varies with labor markets at the time of release. Additionally, the impacts of this policy are of direct interest to policy-makers. I use a difference-in-differences identification strategy to estimate the effect of Ban the Box hiring laws on misdemeanor defendants, felony defendants, and released returning prisoners. BTB policies improve labor market outcomes and lower recidivism for misdemeanor defendants. Evidence is more mixed for felony defendants and people released from prison into jurisdictions that have adopted BTB policies, with some imprecise and inconsistent support

¹Several representative studies include Waldfogel (1994); Grogger (1995); Nagin and Waldfogel (1998); Lalonde and Cho (2008); Kling (2006); Pettit and Lyons (2007); Harding et al. (2018); Mueller-Smith and Schnepel (2021); Manudeep, Scott, and Waddell (2020); Dobbie, Goldin, and Yang (2018); Pager (2003); Pager, Bonikowski, and Western (2009); Mueller-Smith (2015); Pyle (2023).

for improved labor market outcomes and lowered recidivism rates.²

Stronger labor markets may influence recidivism and criminal activity through several channels. One mechanism is the changed employment opportunities for those potentially involved in criminal behavior. With sufficient labor demand, employers may be willing to hire from populations they would not otherwise consider. However, another potential channel is the employment opportunities and resources for the support and social network of at-risk populations. Alternatively, improved labor markets may allow for additional offending opportunities as there are now additional resources available as the target of crime. There may also be more interpersonal contact during economic booms, generating more opportunities for criminal interactions. Other features, like the salience and cost of observing criminal records, may also impact job matching rates. This paper presents evidence that economic booms impact crime through channels other than simply employment available for at-risk populations. This suggests more work should be focused on understanding these mechanisms.

1.1 Background

Most canonical economic models explicitly or implicitly construct a trade-off between licit and illicit work (Becker, 1968; Ehrlich, 1973). Empirically, a recent review summarizing the literature relating to violent crime, Ludwig and Schnepel (2024), suggests that good jobs may discourage property crime, but the evidence on serious violent crime is less promising. Research has suggested that transferring resources to at-risk populations can reduce offending, but context and implementation matter.³ The current paper focuses on overall reoffending and cannot break out impacts by crime type. Because most crimes are property crimes (about 80%), the design in this study will primarily add to our understanding of the relationship between jobs and overall criminal behavior and abstracts from the social cost of the underlying offense.

Several studies, including this one, take a macro approach to identification, studying area-level aggregate measures of labor markets, income, and crime. Bushway, Cook, and Phillips (2013) documents a correlation between recession and income-motivated crimes (like theft crimes) but finds limited evidence of a violent crime business cycle. A series of papers studying local labor market shocks finds that unemployment generated by changes to oil

²One potential explanation for this could be the penalties faced immediately upon conviction or more visible prison record.

³Additional consistent surveys of the relationship between income and crime can be found in Mustard (2010) and Bushway (2011). Work relating to social support and crime includes: Tuttle (2019); Watson, Guettabi, and Reimer (2020); Mueller-Smith et al. (2023); Deshpande and Mueller-Smith (2022); Jacob, Kapustin, and Ludwig (2015).

prices, military base closures, and import price changes tends to increase criminal activity, although it is hard to say the precise mechanism at play (Raphael and Winter-Ebmer, 2001; Gould, Weinberg, and Mustard, 2002; Lin, 2008; Prescott and Pyle, 2019).

Most related to the analysis in this paper are several studies that document increased hiring during the quarter of release from prison in industries that often employ workers with criminal records decreases recidivism rates. Early work in this area showed the state of the labor market upon reentry had mixed impacts on recidivism (Raphael and Weiman, 2007). However, Yang (2017) and Schnepel (2018) use administrative data on releases from prison and unconnected administrative data on local hiring to show that increased hiring and wages in criminal record-friendly industries lower recidivism.⁴ In this paper, I present evidence that strong demand is working through mechanisms beyond the formal employment of returning citizens.

Along with business cycle variation, policies governing hiring job-seekers with criminal histories have changed. One high-profile change is the introduction of Ban the Box. The federal government, 36 states, and over 150 cities and counties have implemented versions of BTB policies. While BTB policies have some implementation variation, a common feature is that employers must delay inquiring about criminal records until later in the hiring process, often after they have made a conditional job offer.

Ban the Box policies are often implemented intending to address the low employment and high recidivism rates observed amongst people with criminal records. Whether the intended effect is achieved has been debated and is the inquiry of this paper. By removing conviction inquiries from applications, BTB proponents hoped to increase applications from job seekers with convictions and encourage employers to make individualized assessments rather than rely on categorical exclusions of these applicants, as BTB allows for an applicant to present evidence of rehabilitation and review background checks. Critics of the policy point out that it may fail to help people with criminal records obtain jobs as it does little to change the underlying economics of hiring a person with a criminal record.

A recent high-quality literature review of BTB policies concludes that “there is some evidence that these laws have the intended impact in the public sector but little evidence of any benefits regarding access to private-sector jobs. To be sure, the body of research addressing this specific question is small. Additional research on this question using different locations and, in the best case, administrative data would certainly be useful given the ubiquity of BTB laws and some of the unintended consequences.” (Raphael, 2021)

⁴People released into labor markets with relatively higher minimum wages had lower recidivism rates (Agan and Makowsky, 2023), although minimum wage increases may increase for some subpopulations like young, less-educated men (Fone, Sabia, and Cesur, 2023).

This paper heeds the call for additional work by analyzing newly publicly available administrative data (Finlay, Mueller-Smith, and the CJARS team, 2024). While many studies have focused on BTB-induced discrimination, there are only a few papers that seek to measure the direct employment and recidivism effects of BTB policies on people with criminal records. However, these few papers typically relied on either résumé correspondence approaches, administrative data from a single state, or data from surveys with relatively small samples that rely on self-reported (and potentially selected) information. This paper analyzes high-quality administrative data across dozens of BTB policies, allowing the current analysis to have several advantages over the correspondence approach (this paper measures actual employment changes instead of callbacks), survey-based approaches (this paper uses administrative data with larger samples and no reliance on self-reporting), and previous administrative data approaches (this paper studies multiple jurisdictions allowing for more control jurisdictions and various implementations of the policy).

One careful and influential paper using a correspondence audit approach finds that BTB may help applicants with criminal records, at least at the callback stage of the application process (Agan and Starr, 2017). It also suggests that it may harm Black applicants without a criminal record (and it may be near neutral to slightly harmful for white applicants without a criminal record). However, translating callback rates to employment outcomes requires substantial assumptions, and a gap in callback rates between groups can support a wide range of wage and unemployment gaps depending on modeling assumptions and parameterizations of important search-theoretic parameters (Rosen, 1997; Fryer, Pager, and Spenkuch, 2013; Lanning, 2013).

Subsequent work has sought to measure the impact of BTB on wage and employment outcomes rather than callback rates. For instance, Craigie (2020) uses data from the National Longitudinal Survey of Youth (NLSY) and a difference-in-differences estimation strategy. It finds increases in public employment rates, but is based on the self-reported outcomes of relatively few people. Ballance, Clifford, and Shoag (2020) finds that employment rates increase in high crime rate census tracts relative to lower crime rate tracts after BTB policies are enacted but cannot directly link employment rates to criminal records.

Two additional studies have the advantage of using high-quality administrative data covering the population of interest (Jackson and Zhao, 2017; Rose, 2021). Jackson and Zhao (2017) studies reforms in Massachusetts but faces challenges with generating a comparison control group. Rose (2021) studies a BTB policy enacted in Seattle using other Washington counties as controls. In the most comparable identification strategy to the one employed in the present paper, Rose (2021) finds noisy employment increases after Seattle's private sector BTB law for the recently released population between 1 and 2.5 p.p. (percentage points),

with some questions about parallel trends being satisfied in the specification yielding an estimate of 2.5 percentage points. Earnings appear to increase by about \$125 per quarter. I build upon this previous work by expanding the studied sample to include those with misdemeanor charges (a previously understudied population, who I find have relatively large benefits from BTB policies) and including more policy changes.

Studying a longer and more recent time frame may be important here as policies evolved. For instance, in April 2012, the Equal Employment Opportunity Commission (EEOC) issued guidance that broad hiring bans and inappropriate use of criminal records in the hiring process could be violations of Title VII. These failure-to-hire claims are challenging to enforce for various reasons, including difficulty in proving that an employer chose not to hire an applicant due to a criminal record. Work analyzing EEOC claims suggests that BTB policies may allow more effective enforcement of these failure-to-hire claims U.S. Equal Employment Opportunity Commission (2023). A second look at BTB policies, especially in the current environment.

Many papers analyzed the impact of BTB on statistical discrimination and found evidence that the policy results in employers hiring fewer young Black and Hispanic men (Doleac and Hansen, 2020; Agan and Starr, 2017). Recent work from Burton and Wasser (2022) and Kaestner and Wang (2024) has suggested that the amount of statistical discrimination generated by BTB may have been smaller than previously understood. I do not address the potentially important unintended consequences of Ban the Box in this paper, focusing instead on measuring the understudied but important intended consequences.

2 Materials and Methods

2.1 Data

This paper relies on several data sources. First, the analysis relies on newly released data from the Criminal Justice Administrative Records System (CJARS) via the Justice Outcome Explorer (Finlay, Mueller-Smith, and the CJARS team, 2024). CJARS collects and harmonizes data from hundreds of justice agencies, including prisons and courts. This data on justice involvement is then linked to employment and other outcomes through a partnership with the Census Bureau. The Justice Outcomes Explorer (JOE) allows for analysis at geographies as fine as the county level. However, the analysis in this paper focuses on the commuting zone level of aggregation to better capture labor markets. This data allows us to overcome many of the data challenges faced in this literature. However, one limitation generated by CJARS JOE data is that analysis conducted for subgroups (for instance, disaggregating the sample by race and age) requires using employment data pooled at the

biannual level, which will impose substantial measurement issues, biasing these estimates towards zero and overstating precision. This limitation does not impact the main findings of this paper, which focus on impacts across the full population in order to avoid this cross-year aggregation. Results using the noisier pooled data focusing on subgroups are only presented in the Appendix and should be interpreted with caution. These results may be useful in thinking about cross-group comparisons, but point estimates and standard errors should be appropriately interpreted. A second limitation of the data relates to the frequency at which measures are observed (yearly, from tax records).⁵ A third limitation is how geographies are defined. Event geographies are generated based on the county of sentencing, which is then aggregated to the commuting zone level. This measure is a rough proxy for the county of residence. The combination of these features suggests a bias towards zero in the following analysis.

Data coverage limitations are imposed by relying on the Finlay, Mueller-Smith, and the CJARS team (2024) data. Not all commuting zones are covered across all criminal justice domains for all time periods. To account for these coverage questions, I generate three samples of interest: a “misdemeanor coverage” sample, a “felony coverage” sample, and an “incarceration coverage” sample.⁶ Cohort coverage runs from 2005 to 2018 to allow for consistent coverage.

The following data sets are available at the county level. I aggregate to 2000 Census-designated commuting zones using the CJARS commuting zone crosswalk. Following Schnepel (2018) and Yang (2017), I use labor market data from the Quarterly Work Force Indicator (QWI) data generated from the Longitudinal Employer-Household Dynamics (LEHD). This data is important for generating high-quality proxies of labor demand. QWI data is built from state unemployment insurance records, which will capture most formal employment relationships, although they may miss important informal employment relationships. I collapse this data to the commuting-zone year level. Population data is downloaded from the National Institute for Health’s Surveillance, Epidemiology, and End Results (SEER) Program county population files. I build employment, unemployment, and labor force data from the Bureau of Labor Statistics Local Area Unemployment Statistics (LAUS).

I follow Doleac and Hansen (2020) to code Ban the Box treatment. I code a commuting zone as having a BTB policy in place if any jurisdiction has a BTB policy within the

⁵Tax records may not capture unreported informal work, which may be important for people with criminal records.

⁶States with misdemeanor coverage are: (AZ, FL, MD, MI, MN, NC, ND, OK, OR, PA, TX, VA, WI: SC is available only for recidivism statistics), felony coverage states include all misdemeanor states but also add AR (only for recidivism) and NJ. States in the incarceration sample are: (AZ, FL, IL, KS, MN, NE, NV, NM, NY, NC, OK, PA, TX, WI: CO, NV, and NM are only available for recidivism statistics and IN is only available for earnings statistics).

commuting zone (any of the following will qualify: state, county, or city-level policy). I use Ban the Box policy adoption dates as described in Avery and Hernandez (2021). To maximize power, I do not distinguish between the types of BTB policies in place. Results distinguishing between BTB laws focused on private versus public employers are available in Appendix Table A13. By the end of the sample period considered, most jurisdictions are covered by some Ban the Box policy.

2.2 Empirical Strategy

This paper uses two sources of quasi-random variation to study labor markets for people with criminal justice involvement: (1) local labor market health during year of release from prison and (2) BTB adoption.

The first strategy relies on the plausibly exogenous state of the labor market during the year a person is released from prison. My empirical specification is most closely related to Schnepel (2018), in that I measure slack in labor supply as well as measures of labor market demand during the period of release. Results are robust to including a lagged dependent variable.⁷ While entry to prison and offending choices are likely endogenous with respect to the labor market, the state of the labor market at the time a person has finished their sentence is plausibly random (Yang, 2017; Schnepel, 2018). Following the literature, I estimate the following equation:

$$\text{Outcome}_{ct} = \alpha + \beta^k \text{New Hires}_{ct}^{ek} + Z'_{ct}\Gamma + X'_{ct}\Pi + \tau_t + \phi_c + \epsilon_{ct} \quad (1)$$

Fixed effects controlling for commuting zone (ϕ_c) and year of release (τ_t) are included in each specification. Commuting zone characteristics (Z_{ct}) include the log transform of the number of released prisoners in a cohort and the unemployment rate in the commuting zone. This also includes controls for a policy that might alter the demand for workers with records, Ban the Box. Some specifications include information on labor market outcomes for the released cohort (X_{ct}) to assess whether the condition of the labor market upon release has an effect after netting out average labor market outcomes. Two sets of outcomes are considered: recidivism outcomes (prison re-entry and felony convictions) and employment outcomes (earning above the poverty rate, any W2 filing, earnings, and the log of earnings). Outcomes over different horizons are considered: one year after release and three years after release.

Following Schnepel (2018), the measure of labor demand used is the number of new non-

⁷Relative to prior work, I observe hiring at the yearly frequency relative to the quarterly frequency, which may bias results towards 0.

recalled male workers hired of education group e and industry k as a share of the commuting zones population.⁸ I split the sample into three education groups: those with high school or less education, those with some college or more, and those who are under 25 (for whom LEHD data does not measure education). Manufacturing and construction are the two industries that have been the most important proxies for the types of work available to returning citizens. I split new hires with high school or less education into those hired into the manufacturing or construction industries and those hired into all other industries. These hires are then scaled as a share of the commuting zone's population (measured in thousands).

I then assess BTB implementation. Following previous studies of Ban the Box adoption, I use a difference-in-differences approach at the commuting zone level. The impact of Ban the Box reforms may vary across time of adoption and labor market tightness (that is, it likely has heterogeneous treatment effects), and states implemented the reform over several years (staggered timing). In the presence of these two features, a standard statistical approach, two-way fixed effects (TWFE), will not yield estimates of the causal relationship of interest (De Chaisemartin and d'Haultfoeuille, 2020; Goodman-Bacon, 2021; Callaway and Sant'Anna, 2021; Gardner, 2022). To account for this, I follow Gardner (2022) and implement an imputation estimation procedure. This first stage predicts counterfactual outcomes in all periods and residualizes the observed outcome. To do so, untreated (or not-yet-treated) observations are used to estimate each coefficient (except for the treatment). Then, the residualized outcomes are regressed on BTB reform-either indicators for years relative to reform enactment for the event studies or an indicator for before/after reform for the overall difference-in-differences estimate, and the standard errors are adjusted to account for the imputation. However, results are similar using the approach found in Callaway and Sant'Anna (2021). These findings are reassuring, as the two approaches rely on slightly different parallel trend assumptions.

In the baseline BTB specifications, I do not include covariate controls beyond commuting zone and year-fixed effects, but the results are robust to the inclusion of controls for local labor force participation rates, sector employment levels and flows, wage rates, size of release cohort, and criminal charge rate (Appendix Table A10).

⁸Results are similar if recalled workers are included or if both male and female hiring are included.

3 Results and Discussion

3.1 Labor market demand and slack at time of release

First, I build on the findings of Schnepel (2018) and Yang (2017) to understand the impact of labor demand at the time of prison release. I document several new facts: first, being released into a commuting zone with higher than usual unemployment rates generates lower recidivism and has a noisy but predictably negative relationship with labor market attachment. Consistent with previous findings, labor demand, as proxied through construction and manufacturing low-education hiring, is associated with short and medium-run decreases in recidivism rates. In new findings, I show higher demand also results in increased short-run employment returning prisoners, although this effect does not persist to a 3-year follow-up. Perhaps surprisingly, the demand-induced reduction in recidivism persists even after directly controlling for employment outcomes of the released cohort.

Given the novel nature of the data, it is helpful to document some basic descriptive facts. To do this, I show simple correlations with labor market outcomes of released cohorts in binned scatter plots. Figure A1 shows the relationship between the share of a cohort of returning citizens earning above the poverty line in the year following release and various measures of the labor market. Each panel nets out the commuting zone and year of release fixed effects. It is worth highlighting that a low share of this group earns above the poverty line, typically ranging between 14% and 24%. Panel B shows the persistence of labor market performance from year to year by comparing the current cohort's labor market performance (earning above the poverty line) with the previous year's cohort labor market outcome. This relationship may be driven by persistent labor demand conditions but may also be driven by successful programming within the prison, peer effects, or countless other features. Panel C-F shows several proxies for labor demand (employment-to-population ratio, employed as a share of labor force participants, hiring in industries and education groups) in the commuting zone-year of release. As a commuting zone employs a higher proportion of its population than usual, more people find jobs that pay above the poverty line in the first year after being released from prison.

I repeat the same analysis shown in Figure A1 in Figure A2, but I now substitute recidivism, defined as prison reentry within one year, in place of earning above the poverty on the y-axis. The scatters do not show an obvious relationship across the various measures between the state of the labor market upon release and prison reentry. Panels A and B suggest that cohorts with better labor market outcomes have lower recidivism rates.⁹ Panels C, E, and F

⁹Panel B shows that it is not just the mechanical relationship of incapacitation on earnings driving the relationship. The “mechanical effect” is the inability to work generated by incarceration. Groups with higher

show little correlation between several local labor market measures and recidivism, at least in the raw data. Somewhat surprisingly, Panel D shows that people released into commuting zones with higher employment rates (lower unemployment rates) have higher rates of prison reentry recidivism.

Table 1 presents a regression analysis. All specifications include commuting zone and year of release fixed effects. Panel A studies the relationship between recidivism and labor market indicators. Columns 1-3 focus on recidivism, which is defined as reincarceration. Columns 4-6 restrict the sample to commuting zones covered in both the “prison” and “felony” samples and define recidivism as the share of the released cohort charged with a new felony. The results are similar across samples and outcome measures. Panel B replaces the dependent variable with various cohort labor market outcomes. Given sample coverage, the studied population varies slightly by column. Results restricting to the sample with full coverage across all dimensions show near identical results (Appendix Tables A1 and A2).

First, I control for whether the jurisdiction has an active Ban the Box policy. People released into commuting zones with BTB policies had recidivism rates between 1 and 2 p.p. lower than commuting zones without a BTB policy in place. Prisoners released into jurisdictions with BTB policies also have better employment outcomes (1.2 p.p. higher rates of earning above the poverty line, 1.3 p.p. higher W2 filing rates, \$486 (5.5%) higher earnings). Analysis in the subsequent subsection provides reasons to be cautious about a causal interpretation of this two-way fixed effects approach. Larger release cohorts appear to have worse outcomes: a 10% increase in cohort size is associated with 2.5 to 2.8% higher reincarceration rates. However, estimates for new felony charges range from 0.6% to 2% but are no longer statistically significant at conventional levels. Larger cohorts also have significantly worse employment outcomes, with a 10% increase in size associated with a 3.8 p.p. decrease in the share earning above the poverty line, a 6.2 p.p. decrease in W2 filing, and a \$1,255 decline in earnings (22%). Cohort size is an important confounder to control for. However, it is difficult to be confident in any causal interpretations as we cannot distinguish what features of cohort size drive this result (for instance, it could be compositional differences or increased supply of competing job seekers).

Measures of local labor market conditions at the time of release are plausibly exogenous and can be interpreted causally (Yang, 2017; Schnepel, 2018). I study several measures of the job markets facing the released cohort that fall into two broad categories: the local unemployment rate and hiring flows (which I then further split by education and industry). Interestingly, increases in local unemployment seem to decrease recidivism rates. The relationship between local unemployment, hiring, and recidivism rates is more attenuated and

rates of incarceration will be unable to earn money during the incarceration spell.

less consistent. Hiring flows are of primary interest in this analysis, as previous research has suggested this is likely the most critical measure of labor market opportunities for returning citizens. A one standard deviation increase (.0017) in low-education construction and manufacturing new hiring as a share of the commuting zones population in the year of the cohort's release causes 1-year reincarceration recidivism declines of about 1.1 p.p. and felony recidivism reductions of about 1.4 p.p. To put these numbers in context, about 14% of released prisoners return to prison within a year, and about 16% have a new felony charge.

Evidence for recidivism reduction caused by low-education hiring in other industries is weaker. A one standard deviation increase in low-education hiring in industries other than manufacturing or construction (.0042) generates an imprecisely estimated (and not statistically significant) 1.2-1.6 p.p. decrease in prison reentry and a statistically significant 2.1 p.p. decline in new felony charges. This hiring causes increased average earnings for the released cohort, although the evidence is more substantial that it increases wages (the intensive margin) than increases labor force attachment (the extensive margin). Specifically, a one standard deviation increase in low-education manufacturing and construction hiring at the time of release increases the share of the released cohort above the poverty line by 1.1 p.p, a statistically insignificant increase in any W2 filing of 0.9, and an increase in earnings of \$428 (3.8%). There is little evidence that the overall local hiring of workers with education beyond a high school degree impacts either the hiring rates of returning prisoners or recidivism. Columns 2 and 5 drop the hiring variables and instead include the cohort's employment outcomes in the first year after release. Of the three measures of labor market attachment, earning above the poverty line is more predictive of recidivism than any W2 filing or wages.

Columns 3 and 6 include both the hiring cohort's employment outcomes and local aggregate hiring rates. The coefficient on low-education construction and manufacturing hiring is almost unchanged after including actual employment outcomes, suggesting that this measure of demand may be influencing recidivism outcomes through channels other than that cohort's employment outcomes.

Table 2 repeats the exercise of Table 1, now looking at outcomes over the three years following release from prison. The three-year follow-up period patterns are broadly consistent with the one-year outcomes. Again, BTB policies and unemployment rate in the year of release seem to have lasting reductions in prison reentry and new felony charges. The unemployment rate at the time of release no longer impacts employment outcomes at the 3-year horizon. Larger cohorts have higher recidivism rates and worse employment outcomes. Construction and manufacturing hiring in the year of release remains strongly predictive of lower recidivism rates at the three-year window. However, this hiring measure no longer pre-

dicts employment outcomes over the three-year window. This time window raises the same questions as the one-year follow-up hiring: labor demand generates lower recidivism, but not (or at least not only) through increasing employment outcomes of the released cohort.

3.2 Ban the Box adoption

Tables 1 and 2 present some suggestive evidence that Ban the Box policies are at least associated with lower recidivism rates and improved employment outcomes in a TWFE approach. While the two-way fixed-effects approach is helpful in understanding the impact of labor market conditions on release cohort employment and recidivism, a more appropriate strategy for assessing the effect of BTB policies requires careful evaluation of parallel trends with an appropriate econometric technique.

Figures 1 and 2 show event studies using the Gardner (2022) imputation approach to difference-in-differences. Table 3 considers point estimates for each outcome and cohort. Results using Callaway and Sant'Anna (2021) are available in Appendix Figures A5, A6, and A7 and Appendix Tables A9 and A8.¹⁰ 7 outcomes are considered: 3 employment measures (earning above poverty, any W2 filing, W2 wages), mortality, and three recidivism measures (a new felony charge, a new misdemeanor charge, or a new prison entry). To avoid challenges of partial treatment in the follow-up period, the analysis focuses on these outcomes in 1-year follow-up periods for three different samples: those with a misdemeanor charge in a given year, those with a felony charge, and those released from prison.

The misdemeanor charge sample has flat pre-trends centered around 0 for each outcome of interest (Figure 1, Table 3 Panel A). Before adopting Ban the Box policies, being charged with a misdemeanor generated similar outcomes across jurisdictions that had yet to reform and jurisdictions that would never reform. Similar patterns in the pre-treatment period are often considered as evidence that the parallel trend assumption is more likely to be satisfied. After a commuting zone adopts a BTB policy, the proportion of misdemeanor defendants who earn above the poverty line in the year following the charge increases 3.1 p.p. (about 10%), the share filing any W2 3.3 p.p. (about 5%), and W2 wage increases of \$1,183 (about 10%). Felony recidivism also declines, dropping about 1.5 p.p.. There appears to be limited impact on future misdemeanor charges. There are modest reductions in mortality as well (.09 p.p.). Results using Callaway and Sant'Anna (2021) are qualitatively similar, with clear improvements across each labor market measure, recidivism, and mortality, although the effects are smaller (Appendix Figure A5 and Appendix Table A3).

Ban the Box appears to have a more minor impact on the sample of felony defendants.

¹⁰The most important difference in these approaches in the context of this analysis relate to parallel trends assumption, with imputation methods relying on a longer pre-period average in the comparisons.

Panel B and Figure A3 show relatively modest (if any) improvements to labor market outcomes, with only the share earning above poverty and reductions in new felony charges significantly different than 0. Results using Callaway and Sant'Anna (2021) are similar, with only the reduction in new felony charges significantly different from 0. As discussed later in the section, controls may be particularly important for the felony defendant sample. Once controls are included (as shown in Appendix Table A10 Panel B), there are statistically significant improvements in labor market outcomes and no changes to recidivism. This pattern may be driven by the longer nature of felony trials and the higher proportion of the sample that is incapacitated by imprisonment and unable to be impacted by changes to employer screening in the year after they are charged.

Evidence on the impact of BTB on people returning from prison is more mixed (Figure A4, Table 3 Panel C). Only the extensive margin (any W2 filing) and reincarceration show any statistically significant evidence of improvement after enacting BTB. However, mortality and misdemeanor recidivism appear to increase due to BTB policies. Using Callaway and Sant'Anna (2021) (or TWFE as shown in Table 1) suggests that there may be some statistically significant labor market improvements and lower rates of recidivism generated by the BTB policies for this population. Ban the Box may be less effective in altering hiring information available at early stages in the hiring process, as other signs of incarceration may be readily inferred from applications via job histories, references, or other information.

Subgroup heterogeneity is of considerable interest when studying Ban the Box. Previous work has suggested that restrictions on hiring information increase statistical discrimination. Previous audit study evidence suggests that older white people with records may benefit more than younger Black applicants. One challenge with subsample analysis in this data set is that several of the measures (labor market and mortality) are averaged across two-year cohorts. This aggregation generates measurement error. It will also generate standard errors that are too small. Recidivism-based statistics do not face this measurement issue and are thus presented in the main text. However, because the exercise of interest is a comparison across the subgroups, some cautious inferences may be drawn, and so results including these outcomes are included in the Appendix.

Appendix Tables A4 and A5 split the sample by race, age, criminal history, and offense type. There are no large differences across race or previous criminal justice exposure, with Black and white defendants, as well as first-time and repeat defendants, seeing similar increases in employment outcomes and decreases in recidivism. There is some weak evidence that defendants aged 25-40 have larger gains from the policy than young defendants and larger increases for violent and property defendants than drug defendants. Appendix Table A5 breaks out the sample into race by age subgroups. Here again, we see relatively little

evidence of heterogeneity beyond the age gradient observed in Appendix Table A4.¹¹

I implement several robustness tests beyond estimating the difference-in-differences with two approaches. First, in Appendix Table A9, I drop one state at a time from the sample, showing that the results are not only driven by an outlier state. The results are largely stable across this exercise, although Arizona appears to be a relatively high-leverage state. I also show the impact of the reform by adopting cohort in Appendix Table A8, which shows the reform's effect is similar across many adopting jurisdictions.

Several other outcomes are considered in Appendix Tables A11 and A12. There are no systematic changes to the misdemeanor cohort size (the number charged with a misdemeanor each year), the rate of misdemeanor charges, or the share of charged defendants convicted after BTB adoption. However, the felony charged defendants display changes in cohort size and conviction rates around adoption. BTB jurisdictions seem to have lower rates of felony charges (although similar cohort sizes) and a higher share of convicted defendants. There are no changes in cohort size or rate for the released prison population. I also consider several general economic factors in Appendix Table A12. I find no significant changes in commuting zone employment rates, high school or lower education construction hiring (and construction hiring), college hiring, and the wages of high school and college hires. There appear to be some labor force participation and high school hiring increases after jurisdictions adopt BTB. Given these patterns of covariate evolution, I show in Appendix Table A10 that the results are similar, if slightly larger in magnitude, when a rich set of controls is included in the difference-in-differences analysis. These controls include various measures of local labor market health, the cohort size studied, and the share convicted.

While not many jurisdictions also adopt Ban the Box laws impacting purely private employers, several do. This adoption pattern allows for an additional, if slightly underpowered, analysis of this different policy. Appendix Table A13 shows that private BTB laws may have some additional reduction to recidivism and increases to employment for misdemeanor defendants on top of the impact of having any public employer BTB law.¹² Indeed, the size of the coefficient on employment outcomes is about 1.5 times as large as the public version of the law (although many of the estimates are not statistically significant). Recidivism reductions appear to be of similar size. Note that all jurisdictions studied with BTB laws covering private employers also have laws covering public employers.

¹¹I repeat a similar analysis for the incarceration sample in Tables A6 and A7, which shows roughly similar impacts across groups, with some weak evidence of larger impacts for older white released prisoners than other groups.

¹²Results for the released prisoner sample seem to cut the opposite direction.

4 Conclusion

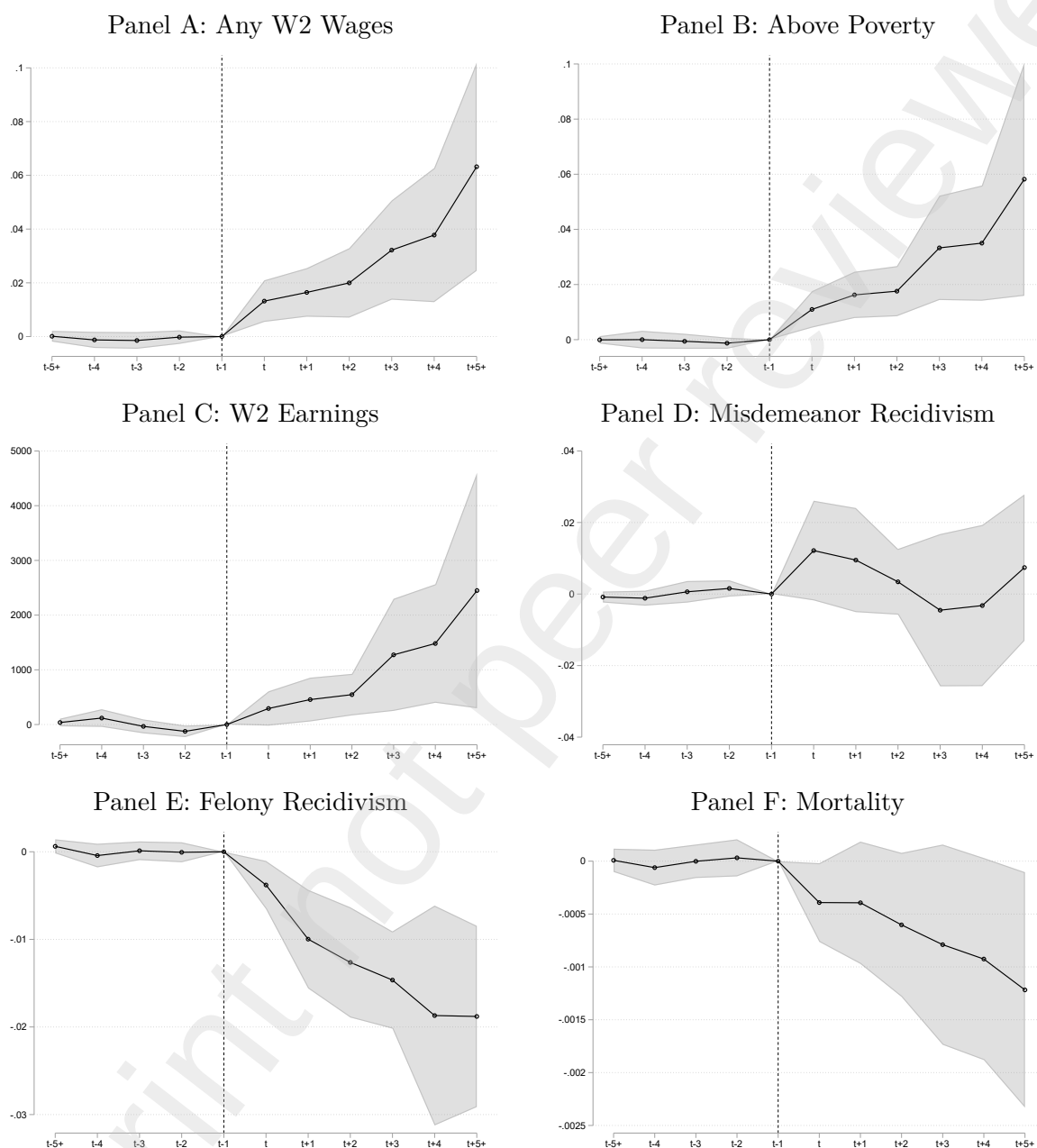
I find that prisoners released into labor markets with higher rates of construction and manufacturing hiring of people with a high school or lower education have significantly lower rates of recidivism. A one standard deviation increase in this type of new hiring in the year of release causes a lower one-year reincarceration recidivism rate of about 1.1 p.p. and a lower felony charge recidivism rate of about 1.4 p.p. A one standard deviation increase in low-education manufacturing and construction hiring at the time of release increases the share of the released cohort above the poverty line by 1.1 p.p., a statistically insignificant increase in any W2 filing of 0.9, and an increase in earnings of \$428 (3.8%). Similar results for recidivism are seen in a 3-year window but not for earnings.

This paper presents new evidence that labor demand is impacting recidivism beyond direct formal employment effects. Controlling directly for release cohort employment outcomes does not significantly reduce the estimated impact of stronger labor market hiring on recidivism outcomes. This pattern suggests that labor market prospects of support networks or other potential channels may be important mechanisms relative to the direct employment of the returning cohort.

Evidence suggests Ban the Box policies impact the labor market and recidivism outcomes for people charged with misdemeanors. Previous papers have been unable to document this impact due to data limitations, and few have focused on those facing misdemeanors. By leveraging the largest data set on workers with criminal justice involvement to date, I show that after enacting BTB policies, the proportion of misdemeanor defendants who earn above the poverty line in the year following the charge increases by 3.1 percentage points (about 10%), the share filing any W2 by 3.3 percentage points (about 5%), and W2 wages increase \$1,183 (about 10%). Felony recidivism rates decline 1.5 p.p. Results for felony defendants and returning prisoners are more mixed. Although they are often similarly signed as the misdemeanor defendant results, they only become consistently significant with the inclusion of controls or with Callaway and Sant'Anna (2021) estimation methodology. This paper does not study statistical discrimination generated by the adoption of the Ban the Box, but policymakers should carefully weigh the evidence of improved outcomes for those involved with the criminal legal system against potential harms generated by statistical discrimination.

5 Figures

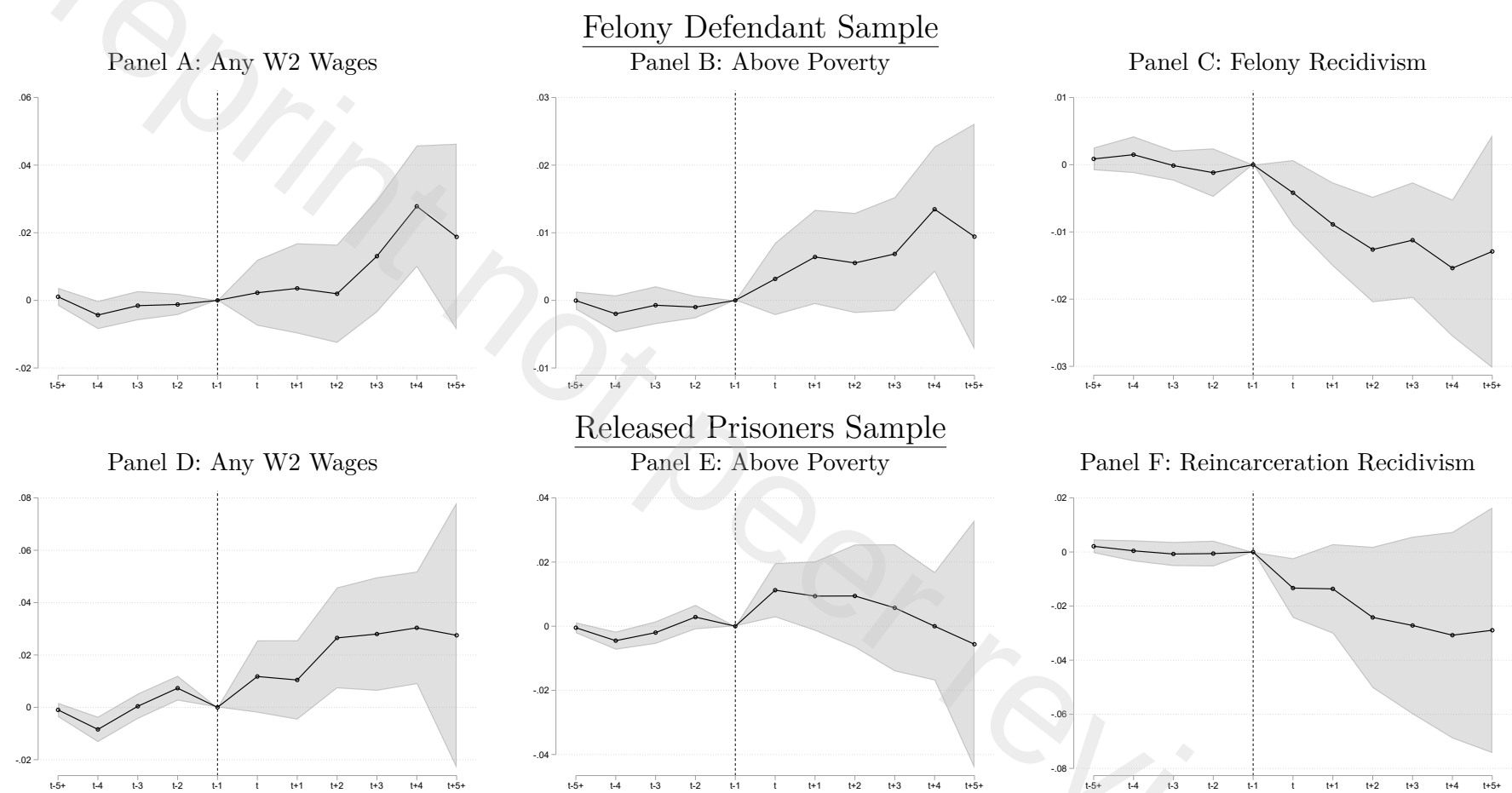
Figure 1: Ban the Box Event Studies: Misdemeanor Defendant Sample



Sources: CJARS JOE 2023Q3 Commuting Zone Data

Notes: This figure reports event study estimates of having any Ban the Box policy in place. Outcomes are measured at the 1-year horizon. Each point represents the interaction of a Ban the Box policy indicator and the corresponding relative year indicator (e.g. $t-2$ indicates the estimate for two years prior to policy implementation). Periods 5 or more years before or after BTB implementation are binned together. Each coefficient estimates the outcome relative to the year prior to the adoption of the policy. Estimates are population-weighted. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the commuting zone level. All estimates follow the approach outlined in Gardner (2022).

Figure 2: Ban the Box Event Studies: Felony Defendant and Released Prisoners Sample



Sources: CJARS JOE 2023Q3 Commuting Zone Data

Notes: This figure reports event study estimates of having any Ban the Box policy in place. Outcomes are measured at the 1-year horizon. Each point represents the interaction of a Ban the Box policy indicator and the corresponding relative year indicator (e.g. t-2 indicates the estimate for two years prior to policy implementation). Periods 5 or more years before or after BTB implementation are binned together. Each coefficient estimates the outcome relative to the year prior to the adoption of the policy. Estimates are population-weighted. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the commuting zone level. All estimates follow the approach outlined in Gardner (2022). Full event studies for each outcome are available in Appendix Figure A3 and Appendix Figure A4.

6 Tables

Table 1: The Effects of Labor Demand At Release on Recidivism and Employment: 1-year Follow-up Results

Panel A: Recidivism

	Re-incarceration			Felony		
	(1)	(2)	(3)	(4)	(5)	(6)
Ban the Box	-0.0178*** (0.00486)	-0.0117** (0.00559)	-0.0111** (0.00519)	-0.0139** (0.00685)	-0.0139** (0.00672)	-0.0111* (0.00633)
Log(Number Released)	0.0258** (0.0103)	0.0249** (0.0115)	0.0279*** (0.0103)	0.0198 (0.0160)	0.00655 (0.0148)	0.0121 (0.0160)
Unemployment Rate	-0.671*** (0.167)	-0.856*** (0.171)	-0.898*** (0.165)	-0.534*** (0.175)	-0.711*** (0.188)	-0.544*** (0.183)
Other low-education hiring	-3.870 (2.368)		-2.801 (2.710)	-5.165** (2.513)		-5.105** (2.408)
Const. and manuf. low-ed. hiring	-6.735*** (2.547)		-6.635** (3.083)	-8.628*** (2.411)		-6.103** (2.570)
High-education hiring	2.588 (2.320)		0.789 (3.641)	3.273 (2.374)		3.212 (2.694)
Under 25 hiring	1.932 (2.226)		5.000* (2.861)	6.354** (2.709)		6.910** (2.816)
Above poverty (1 year)		-0.508*** (0.0826)	-0.510*** (0.0898)		-0.188** (0.0756)	-0.235** (0.0925)
Any W2 wages (1 year)		0.0570 (0.0783)	0.0432 (0.0722)		-0.0348 (0.0839)	-0.0715 (0.0775)
Log(W2 wages) (1 year)		-0.0115 (0.0191)	-0.00893 (0.0175)		-0.0126 (0.0182)	-0.00282 (0.0188)
Observations	3148	2768	2715	1834	1851	1814
Mean	0.148	0.139	0.139	0.159	0.159	0.159

Panel B: Employment

	(1)	(2)	(3)	(4)
	Above Poverty	Any W2	Earnings	Log(Earnings)
Ban the Box	0.0115*** (0.00398)	0.0125** (0.00585)	485.8*** (140.9)	0.0547** (0.0223)
Log(Number Released)	-0.0383*** (0.00620)	-0.0615*** (0.0101)	-1255.4*** (220.5)	-0.215*** (0.0294)
Unemployment Rate	-0.131 (0.0817)	-0.690*** (0.146)	478.9 (3188.0)	-2.339*** (0.396)
Other low-education hiring	0.114 (1.476)	-1.666 (2.645)	42527.2 (48803.2)	-1.757 (7.846)
Const. and manuf. low-ed. hiring	6.349*** (1.868)	4.867 (3.204)	238035.0*** (76871.8)	21.58** (8.761)
High-education hiring	-0.836 (1.407)	1.897 (1.902)	-87085.3 (58071.5)	5.056 (7.303)
Under 25 hiring	2.793** (1.194)	2.845 (2.545)	132838.4*** (50531.5)	7.762 (6.581)
Observations	2713	2713	2713	2713
Mean	0.182	0.497	6596.5	8.861

Sources: CJARS JOE 2023Q3 Commuting Zone Data; LEHD QWI; BLS LAUS; NIH SEER

Notes: Each column represents a separate regression. All regressions include year and commuting zone fixed effects. Hiring variables are divided by CZ population. Estimates are population-weighted. Commuting zone clustered standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < .01$

Table 2: The Effects of Labor Demand At Release on Recidivism and Employment: 3-year Follow-up Results

Panel A: Recidivism

	Re-incarceration			Felony		
	(1)	(2)	(3)	(4)	(5)	(6)
Ban the Box	-0.0161*** (0.00531)	-0.0167*** (0.00595)	-0.0160*** (0.00576)	-0.0155* (0.00924)	-0.0173* (0.00957)	-0.0125 (0.00871)
Log(Number Released)	0.0321*** (0.00888)	0.0342*** (0.00945)	0.0341*** (0.00955)	0.00604 (0.0228)	-0.0139 (0.0218)	-0.00488 (0.0232)
Unemployment Rate	-0.921*** (0.175)	-1.219*** (0.193)	-1.129*** (0.183)	-0.968*** (0.267)	-1.021*** (0.283)	-1.037*** (0.284)
Other low-education hiring	-1.961 (2.410)		-0.757 (2.829)	-10.52** (4.085)		-10.06** (3.944)
Const. and manuf. low-ed. hiring	-4.604** (2.312)		-2.701 (2.861)	-18.64*** (3.896)		-15.33*** (4.031)
High-education hiring	2.889 (2.280)		0.837 (3.222)	7.464* (4.213)		7.090 (4.490)
Under 25 hiring	-0.0956 (1.763)		2.779 (2.273)	8.197** (4.065)		8.334* (4.303)
Above poverty (1 year)		-0.463*** (0.105)	-0.472*** (0.114)		-0.0966 (0.0854)	-0.128 (0.100)
Any W2 wages (1 year)		-0.00106 (0.0668)	-0.00918 (0.0640)		0.0292 (0.114)	-0.0115 (0.103)
Log(W2 wages) (1 year)		-0.0211 (0.0218)	-0.0208 (0.0219)		-0.0620** (0.0254)	-0.0477* (0.0249)
Observations	3002	2619	2572	1693	1707	1674
Mean	0.316	0.307	0.307	0.332	0.332	0.332

Panel B: Employment

	(1)	(2)	(3)	(4)
	Above Poverty	Any W2	Earnings	Log(Earnings)
Ban the Box	0.0108*** (0.00350)	0.0203*** (0.00537)	405.2*** (138.0)	0.0729*** (0.0206)
Log(Number Released)	-0.0284*** (0.00754)	-0.0423*** (0.0122)	-595.3** (255.9)	-0.175*** (0.0409)
Unemployment Rate	-0.00604 (0.103)	0.0649 (0.175)	-1456.3 (3559.2)	-0.388 (0.601)
Other low-education hiring	-2.982** (1.151)	-4.882** (2.059)	-94688.1* (49299.6)	-17.45** (6.840)
Const. and manuf. low-ed. hiring	-0.873 (1.779)	0.220 (3.002)	-50654.1 (70811.2)	-6.532 (10.54)
High-education hiring	1.276 (1.050)	0.615 (1.774)	32645.4 (52327.4)	6.231 (6.232)
Under 25 hiring	0.403 (1.347)	1.240 (2.601)	32754.7 (52733.3)	5.039 (7.885)
Observations	2407	2407	2407	2407
Mean	0.181	0.424	6809.0	8.884

Sources: CJARS JOE 2023Q3 Commuting Zone Data; LEHD QWI; BLS LAUS; NIH SEER

Notes: Each column represents a separate regression. All regressions include year and commuting zone fixed effects. Hiring variables are divided by CZ population. Estimates are population-weighted. Commuting zone clustered standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < .01$

Table 3: Effects of Ban the Box on One Year Outcomes

Panel A: Misdemeanor Defendant Sample

	Above Poverty	Any W2	W2 Wages	Mortality	New Felony	New Misdemeanor
Ban the Box	0.0308*** (0.00921)	0.0331*** (0.00983)	1183.0*** (449.7)	-0.000751** (0.000341)	-0.0147*** (0.00285)	0.00639 (0.00740)
Mean	0.312	0.614	12719.9	0.00941	0.0735	0.169

Panel B: Felony Defendant Sample

	Above Poverty	Any W2	W2 Wages	Mortality	New Felony	New Misdemeanor
Ban the Box	0.00726* (0.00409)	0.0107 (0.00746)	172.9 (128.7)	-0.000161 (0.000445)	-0.0112** (0.00463)	0.0106 (0.0121)
Mean	0.164	0.452	6468.5	0.00983	0.115	0.175

Panel C: Returning Prisoner Sample

	Above Poverty	Any W2	W2 Wages	Mortality	New Felony	New Misdemeanor	Reincarceration
Ban the Box	0.00411 (0.00783)	0.0219** (0.00977)	172.6 (275.3)	0.00144** (0.000711)	-0.0127 (0.0108)	0.0475** (0.0227)	-0.0193* (0.0104)
Mean	0.183	0.500	6645.9	0.0191	0.159	0.178	0.149

Sources: CJARS JOE 2023Q3 Commuting Zone Data

Notes: Each column represents a separate difference in differences regression estimated following Gardner (2022). All regressions include year and commuting zone fixed effects. Estimates are population-weighted. Commuting zone clustered standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < .01$

7 Acknowledgements

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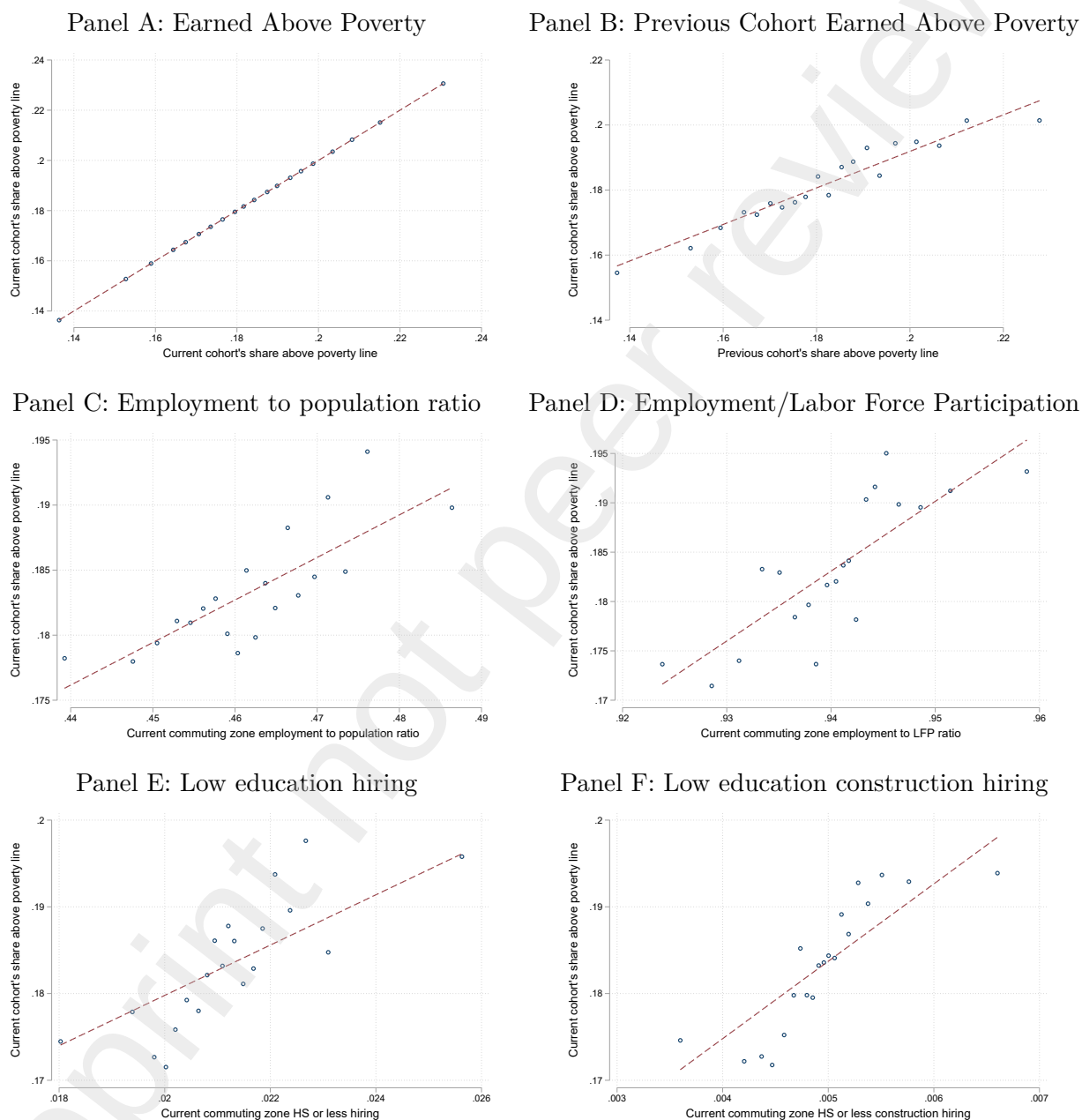
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Appendix: Additional Results

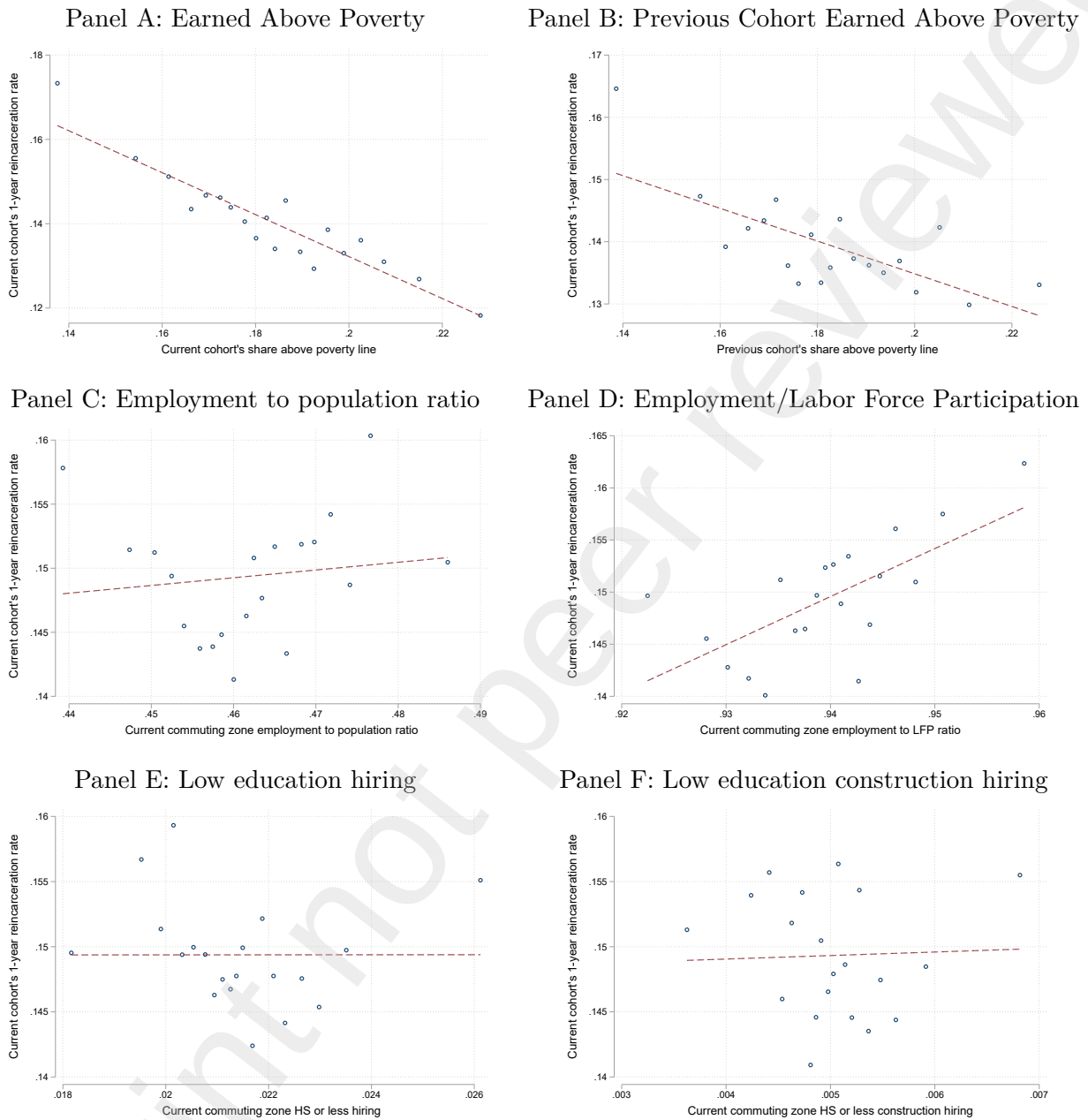
Appendix Figure A1: Relationship between labor markets and returning cohort poverty rates



Sources: CJARS JOE 2023Q3 Commuting Zone Data; LEHD QWI; NIH SEER Population Estimates

Notes: Binscatters control for commuting zone and year of release fixed effects. Data is the “incarceration sample” including cohorts released from prison between 2005 and 2018. Blue dots are binned observations, and the dashed line is a linear fit weighted by cohort size.

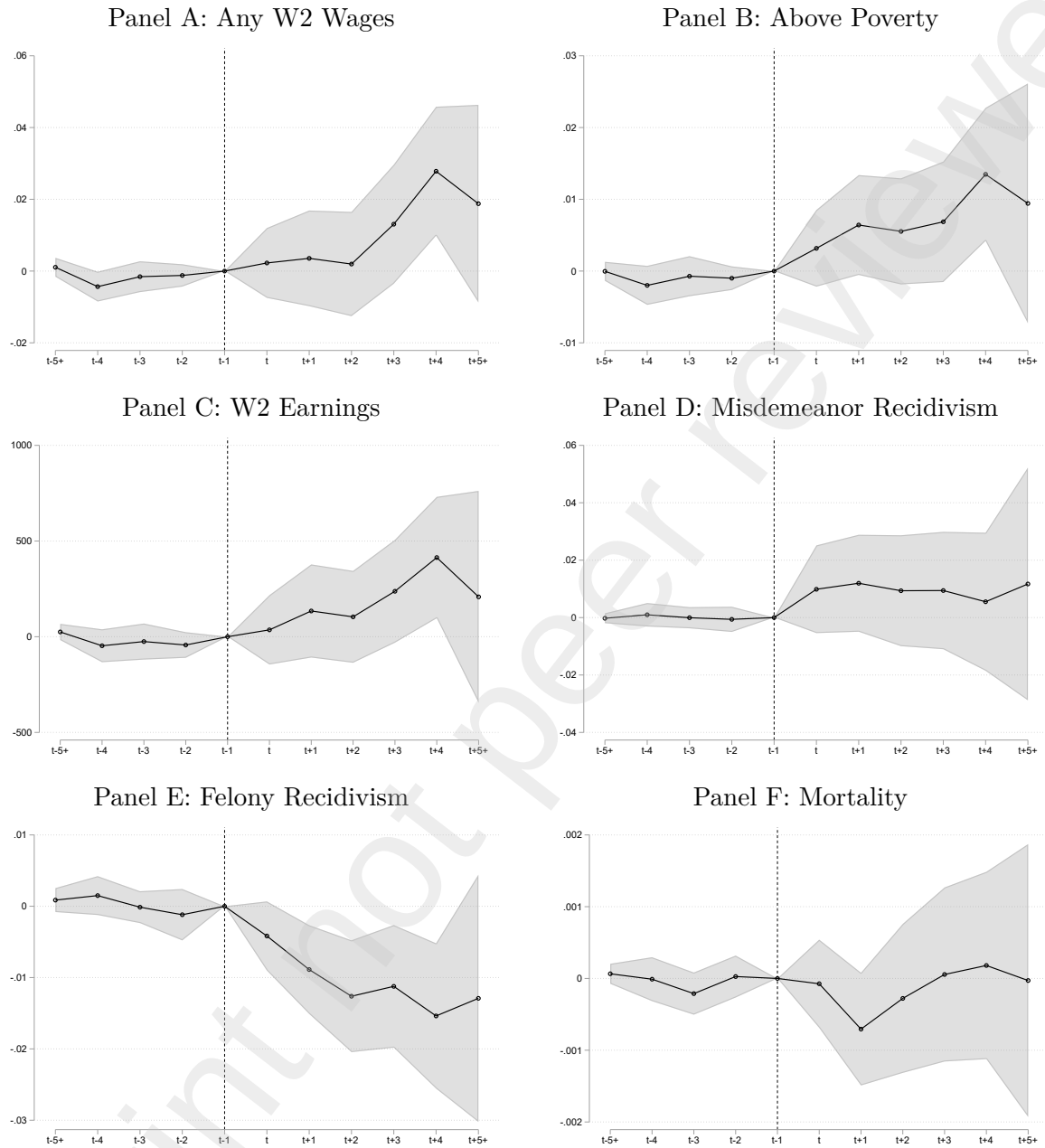
Appendix Figure A2: Relationship between labor markets and crime



Sources: CJARS JOE 2023Q3 Commuting Zone Data; LEHD QWI; NIH SEER Population Estimates

Notes: Binscatters control for commuting zone and year of release fixed effects. Data is the “incarceration sample” including cohorts released from prison between 2005 and 2018. Blue dots are binned observations, and the dashed line is a linear fit weighted by cohort size.

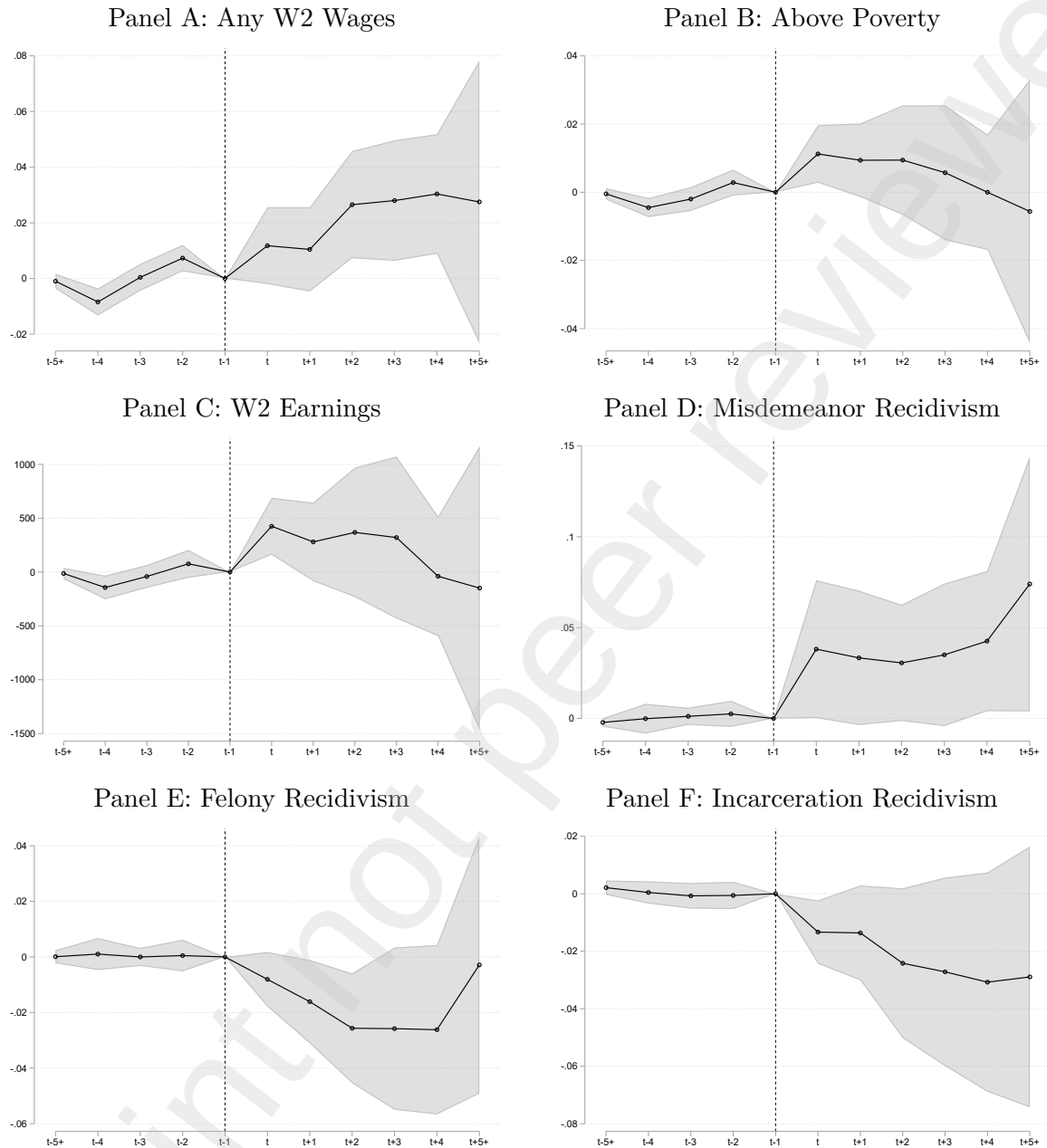
Appendix Figure A3: Ban the Box Event Studies: Felony Defendant Sample



Sources: CJARS JOE 2023Q3 Commuting Zone Data

Notes: This figure reports event study estimates of having any Ban the Box policy in place. Outcomes are measured at the 1-year horizon. Each point represents the interaction of a Ban the Box policy indicator and the corresponding relative year indicator (e.g. t-2 indicates the estimate for two years prior to policy implementation). Periods 5 or more years before or after BTB implementation are binned together. Each coefficient estimates the outcome relative to the year prior to the adoption of the policy. Estimates are population-weighted. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the commuting zone level. All estimates follow the approach outlined in Gardner (2022).

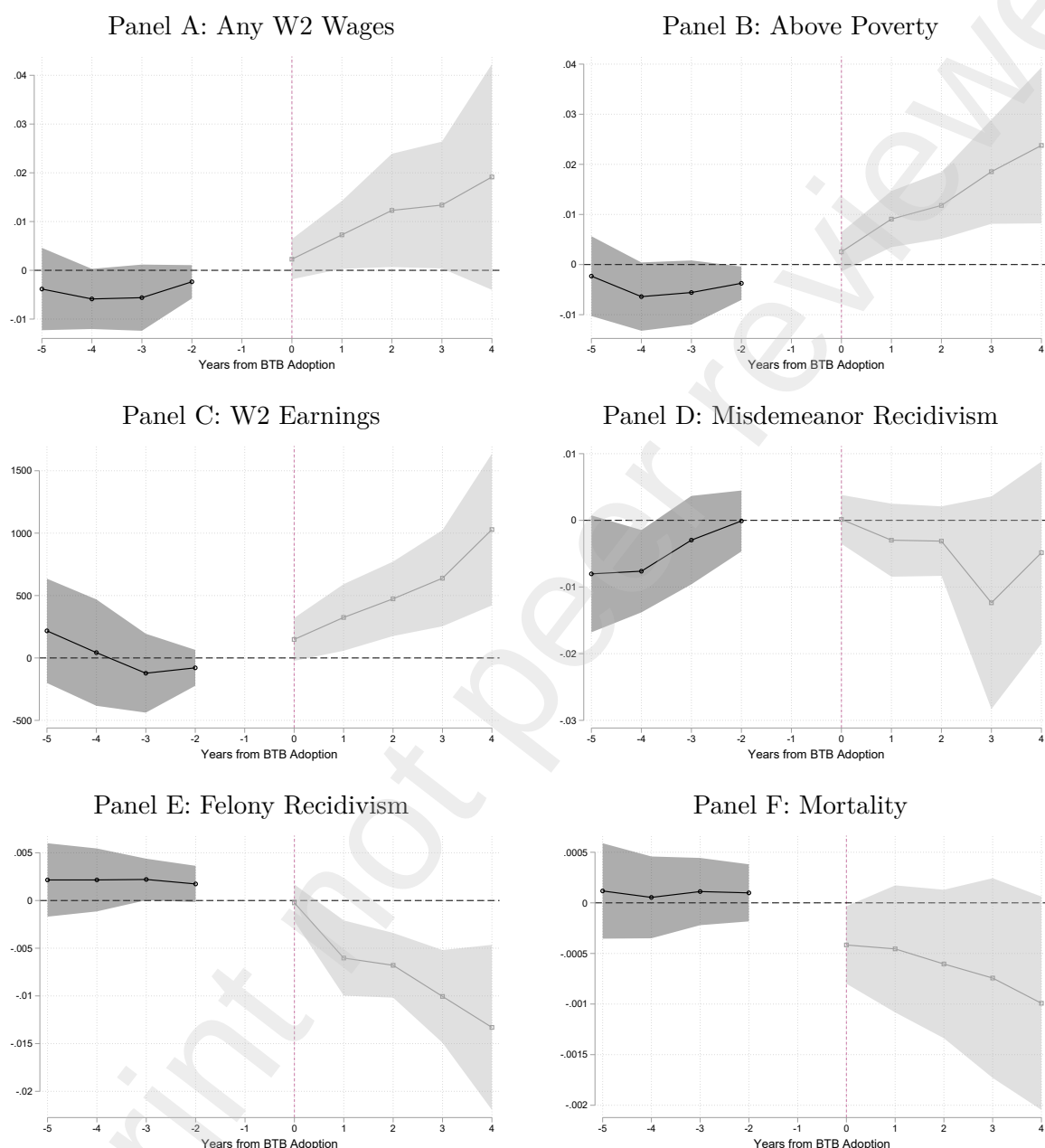
Appendix Figure A4: Ban the Box Event Studies: Returning Prisoner Sample



Sources: CJARS JOE 2023Q3 Commuting Zone Data

Notes: This figure reports event study estimates of having any Ban the Box policy in place. Outcomes are measured at the 1-year horizon. Each point represents the interaction of a Ban the Box policy indicator and the corresponding relative year indicator (e.g. t-2 indicates the estimate for two years prior to policy implementation). Periods 5 or more years before or after BTB implementation are binned together. Each coefficient estimates the outcome relative to the year prior to the adoption of the policy. Estimates are population-weighted. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the commuting zone level. All estimates follow the approach outlined in Gardner (2022).

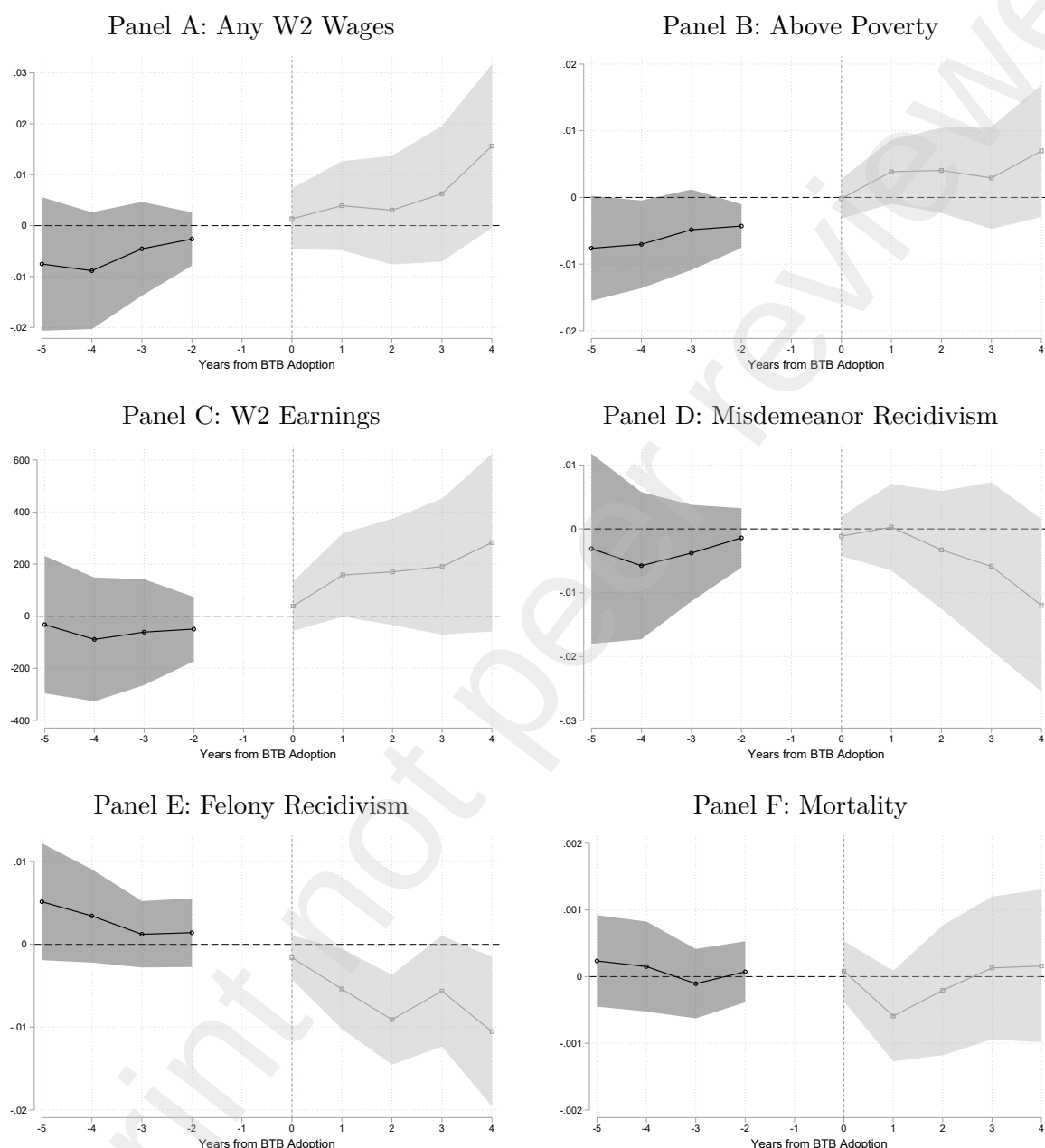
Appendix Figure A5: Ban the Box Event Studies: Misdemeanor Defendant Sample, Estimated Following Callaway and Sant'Anna (2021)



Sources: CJARS JOE 2023Q3 Commuting Zone Data

Notes: This figure reports event study estimates of having any Ban the Box policy in place. Outcomes are measured at the 1-year horizon. Each point represents the interaction of a Ban the Box policy indicator and the corresponding relative year indicator (e.g. t-2 indicates the estimate for two years prior to policy implementation). Periods 6 or more years before or after BTB implementation are omitted. Each coefficient estimates the outcome relative to the year prior to the adoption of the policy. Estimates are population-weighted. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the commuting zone level. All estimates follow the approach outlined in Callaway and Sant'Anna (2021).

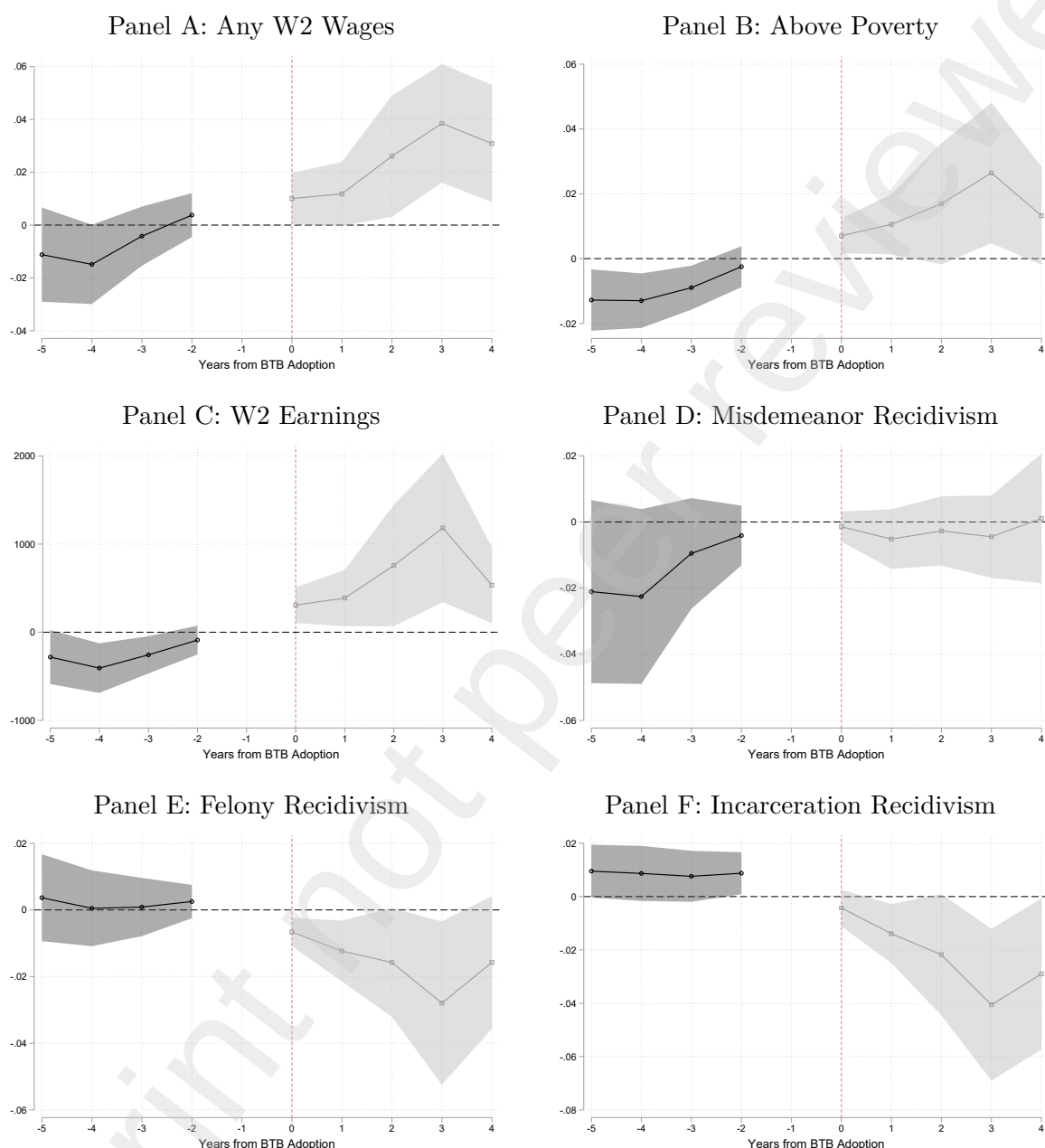
Appendix Figure A6: Ban the Box Event Studies: Felony Defendant Sample, Estimated Following Callaway and Sant'Anna (2021)



Sources: CJARS JOE 2023Q3 Commuting Zone Data

Notes: This figure reports event study estimates of having any Ban the Box policy in place. Outcomes are measured at the 1-year horizon. Each point represents the interaction of a Ban the Box policy indicator and the corresponding relative year indicator (e.g. t-2 indicates the estimate for two years prior to policy implementation). Periods 6 or more years before or after BTB implementation are omitted. Each coefficient estimates the outcome relative to the year prior to the adoption of the policy. Estimates are population-weighted. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the commuting zone level. All estimates follow the approach outlined in Callaway and Sant'Anna (2021).

Appendix Figure A7: Ban the Box Event Studies: Returning Prisoner Sample, Estimated Following Callaway and Sant'Anna (2021)



Sources: CJARS JOE 2023Q3 Commuting Zone Data

Notes: This figure reports event study estimates of having any Ban the Box policy in place. Outcomes are measured at the 1-year horizon. Each point represents the interaction of a Ban the Box policy indicator and the corresponding relative year indicator (e.g. t-2 indicates the estimate for two years prior to policy implementation). Periods 6 or more years before or after BTB implementation are omitted. Each coefficient estimates the outcome relative to the year prior to the adoption of the policy. Estimates are population-weighted. Shaded regions represent 95 percent confidence intervals with standard errors clustered at the commuting zone level. All estimates follow the approach outlined in Callaway and Sant'Anna (2021).

Appendix Table A1: The Effects of Labor Demand At Release on Recidivism: 1-year Follow-up Results, Restricted Sample.

Panel A: Recidivism

	Re-incarceration			Felony		
	(1)	(2)	(3)	(4)	(5)	(6)
Ban the Box	-0.0190*** (0.00570)	-0.0117** (0.00559)	-0.0111** (0.00519)	-0.0134* (0.00681)	-0.0139** (0.00672)	-0.0111* (0.00633)
Log(Number Released)	0.0452*** (0.00890)	0.0249** (0.0116)	0.0279*** (0.0103)	0.0221 (0.0160)	0.00658 (0.0148)	0.0121 (0.0160)
Unemployment Rate	-0.896*** (0.175)	-0.859*** (0.172)	-0.898*** (0.165)	-0.533*** (0.173)	-0.715*** (0.189)	-0.544*** (0.183)
Other low-education hiring	-2.674 (2.621)		-2.801 (2.710)	-5.002** (2.514)		-5.105** (2.408)
Const. and manuf. low-ed. hiring	-8.849*** (2.841)		-6.635** (3.083)	-8.287*** (2.409)		-6.103** (2.570)
High-education hiring	0.706 (2.978)		0.789 (3.641)	3.173 (2.386)		3.212 (2.694)
Under 25 hiring	3.741 (2.540)		5.000* (2.861)	6.230** (2.735)		6.910** (2.816)
Above poverty (1 year)		-0.508*** (0.0827)	-0.510*** (0.0898)		-0.188** (0.0759)	-0.235** (0.0925)
Any W2 wages (1 year)		0.0565 (0.0785)	0.0432 (0.0722)		-0.0353 (0.0842)	-0.0715 (0.0775)
Log(W2 wages) (1 year)		-0.0115 (0.0191)	-0.00893 (0.0175)		-0.0127 (0.0182)	-0.00282 (0.0188)
Observations	2715	2715	2715	1814	1814	1814
Mean	0.139	0.139	0.139	0.159	0.159	0.159

Sources: CJARS JOE 2023Q3 Commuting Zone Data; LEHD QWI; BLS LAUS; NIH SEER

Notes: Each column represents a separate regression. All regressions include year and commuting zone fixed effects. Hiring variables are divided by CZ population. Estimates are population-weighted. Commuting zone clustered standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < .01$

Appendix Table A2: The Effects of Labor Demand At Release on Recidivism: 3-year Follow-up Results, Restricted Sample.

Panel A: Recidivism

	Re-incarceration			Felony		
	(1)	(2)	(3)	(4)	(5)	(6)
Ban the Box	-0.0199*** (0.00616)	-0.0167*** (0.00596)	-0.0160*** (0.00576)	-0.0151 (0.00919)	-0.0173* (0.00957)	-0.0125 (0.00871)
Log(Number Released)	0.0493*** (0.00932)	0.0343*** (0.00948)	0.0341*** (0.00955)	0.00845 (0.0229)	-0.0139 (0.0219)	-0.00488 (0.0232)
Unemployment Rate	-1.112*** (0.195)	-1.222*** (0.194)	-1.129*** (0.183)	-0.978*** (0.262)	-1.026*** (0.285)	-1.037*** (0.284)
Other low-education hiring	-0.841 (2.744)		-0.757 (2.829)	-10.19** (4.103)		-10.06** (3.944)
Const. and manuf. low-ed. hiring	-6.149** (2.543)		-2.701 (2.861)	-18.03*** (3.930)		-15.33*** (4.031)
High-education hiring	1.256 (2.873)		0.837 (3.222)	7.202* (4.255)		7.090 (4.490)
Under 25 hiring	1.500 (1.960)		2.779 (2.273)	7.943* (4.097)		8.334* (4.303)
Above poverty (1 year)		-0.463*** (0.105)	-0.472*** (0.114)		-0.0959 (0.0857)	-0.128 (0.100)
Any W2 wages (1 year)		-0.00144 (0.0670)	-0.00918 (0.0640)		0.0288 (0.114)	-0.0115 (0.103)
Log(W2 wages) (1 year)		-0.0211 (0.0219)	-0.0208 (0.0219)		-0.0622** (0.0254)	-0.0477* (0.0249)
Observations	2572	2572	2572	1674	1674	1674
Mean	0.307	0.307	0.307	0.332	0.332	0.332

Sources: CJARS JOE 2023Q3 Commuting Zone Data; LEHD QWI; BLS LAUS; NIH SEER

Notes: Each column represents a separate regression. All regressions include year and commuting zone fixed effects. Hiring variables are divided by CZ population. Estimates are population-weighted. Commuting zone clustered standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < .01$

Appendix Table A3: Effects of Ban the Box on One Year Outcomes, Estimated Following Callaway and Sant'Anna (2021)

Panel A: Misdemeanor Defendant Sample

	Above Poverty	Any W2	W2 Wages	Mortality	New Felony	New Misdemeanor
Ban the Box	0.0180*** (0.00694)	0.0181** (0.00869)	664.8*** (244.9)	-0.000594* (0.000309)	-0.00844*** (0.00246)	-0.00700* (0.00393)
Mean	0.312	0.614	12719.9	0.00941	0.0735	0.169

Panel B: Felony Defendant Sample

	Above Poverty	Any W2	W2 Wages	Mortality	New Felony	New Misdemeanor
Ban the Box	0.00516 (0.00374)	0.00792 (0.00592)	188.3 (121.3)	-0.0000339 (0.000392)	-0.00958*** (0.00363)	-0.00838 (0.00567)
Mean	0.164	0.452	6468.5	0.00983	0.115	0.175

Panel C: Returning Prisoner Sample

	Above Poverty	Any W2	W2 Wages	Mortality	New Felony	New Misdemeanor	Reincarceration
Ban the Box	0.0167*** (0.00472)	0.0285*** (0.00785)	663.6*** (174.4)	0.00137* (0.000800)	-0.0190*** (0.00577)	-0.00401 (0.00474)	-0.0267*** (0.00747)
Mean	0.183	0.500	6645.9	0.0191	0.159	0.178	0.149

Sources: CJARS JOE 2023Q3 Commuting Zone Data

Notes: Each point estimate represents a separate difference in differences regression estimated following Callaway and Sant'Anna (2021). Estimates are the ATT for all groups across all periods. All regressions include year and commuting zone fixed effects. Estimates are population-weighted. Commuting zone clustered standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < .01$

Appendix Table A4: Effects of Ban the Box on One Year Outcomes by Subgroup:
Misdemeanor Defendant Sample

	Above poverty	Any W2	W2 Wages	Mortality	New Felony	New Misdemeanor
Race						
White	0.0224** (0.0113)	0.0225** (0.00963)	1036.0 (664.2)	0.000418 (0.000424)	-0.0153*** (0.00359)	0.00468 (0.00448)
Mean	0.339	0.634	14549.9	0.0111	0.0674	0.158
Black	0.0210** (0.0103)	0.0230* (0.0127)	396.1 (288.9)	-0.000770*** (0.000266)	-0.0141*** (0.00476)	0.00761 (0.0104)
Mean	0.252	0.581	9351.7	0.00775	0.0959	0.210
Age						
Age 15-24	0.0204* (0.0115)	0.0323*** (0.0125)	640.1* (337.1)	-0.000139 (0.000170)	-0.0155*** (0.00356)	0.0179 (0.0112)
Mean	0.258	0.703	8775.3	0.00475	0.0924	0.206
Age 25-40	0.0358*** (0.0114)	0.0367*** (0.0120)	1167.5** (529.8)	0.000186 (0.000334)	-0.0178*** (0.00328)	0.0109 (0.00905)
Mean	0.340	0.621	13672.8	0.00743	0.0828	0.176
Age 40+	0.0280*** (0.0107)	0.0237** (0.0103)	1491.1* (801.9)	-0.00117* (0.000616)	-0.0124*** (0.00331)	-0.00000577 (0.00781)
Mean	0.275	0.451	13352.0	0.0198	0.0533	0.135
Record						
First Time	0.0368*** (0.0120)	0.0400*** (0.0119)	1733.4** (679.1)	-0.000890* (0.000534)	-0.0166*** (0.00328)	-0.00248 (0.00442)
Mean	0.380	0.688	16226.5	0.00700	0.0340	0.0873
Repeat	0.0451*** (0.0122)	0.0599*** (0.0156)	1481.1*** (414.0)	-0.000404 (0.000512)	-0.0177*** (0.00424)	0.0107 (0.0112)
Mean	0.269	0.563	10527.9	0.0118	0.101	0.215
Crime Type						
Violent	0.0167*** (0.00480)	0.0148** (0.00631)	460.2** (204.1)	-0.000210 (0.000332)	-0.0102*** (0.00280)	0.0121 (0.0102)
Mean	0.287	0.585	11846.6	0.00973	0.0854	0.177
Property	0.0195*** (0.00692)	0.0257*** (0.00984)	344.1 (322.3)	-0.00182*** (0.000470)	-0.0141*** (0.00497)	0.00956 (0.00868)
Mean	0.230	0.557	9170.6	0.0116	0.102	0.191
Drug	0.00887 (0.00690)	0.0140 (0.00992)	272.3 (210.9)	0.000583 (0.000509)	-0.00530 (0.00558)	0.00385 (0.0110)
Mean	0.240	0.588	8646.1	0.0103	0.107	0.201

Sources: CJARS JOE 2023Q3 Commuting Zone Data

Notes: Each point estimate represents a separate difference in differences regression estimated following Gardner (2022) restricted to the population indicated. All regressions include year and commuting zone fixed effects. Gray-shaded columns are subject to additional measurement error due to disclosure pooling rules in CJARS JOE data/ Estimates are population-weighted. Commuting zone clustered standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < .01$

Appendix Table A5: Effects of Ban the Box on One Year Outcomes by Age and Race:
Misdemeanor Defendant Sample

	Above poverty	Any W2	W2 Wages	Mortality	New Felony	New Misdemeanor
White 15-24	0.0199*	0.0270**	593.1	0.000300	-0.0183***	0.0111
	(0.0113)	(0.0107)	(367.0)	(0.000286)	(0.00476)	(0.00764)
Mean	0.302	0.759	10220.2	0.00489	0.0742	0.186
Black 15-24	0.0204	0.0259	582.0	-0.000720**	-0.0142**	0.0225
	(0.0145)	(0.0183)	(465.6)	(0.000366)	(0.00610)	(0.0153)
Mean	0.177	0.631	6108.2	0.00501	0.133	0.260
White 25-40	0.0294**	0.0261**	1165.5	0.00156***	-0.0194***	0.00853
	(0.0142)	(0.0121)	(812.0)	(0.000572)	(0.00434)	(0.00545)
Mean	0.367	0.642	15328.7	0.00912	0.0834	0.172
Black 25-40	0.0312**	0.0375**	738.3**	-0.000403	-0.0156***	0.0139
	(0.0137)	(0.0155)	(345.4)	(0.000303)	(0.00476)	(0.0123)
Mean	0.287	0.597	10316.2	0.00596	0.100	0.213
White 40+	0.0297**	0.0234*	1788.4	-0.000316	-0.0142***	0.000383
	(0.0141)	(0.0122)	(1164.6)	(0.000746)	(0.00342)	(0.00477)
Mean	0.298	0.464	15699.8	0.0226	0.0495	0.125
Black 40+	0.0190*	0.0159	405.6	-0.00123**	-0.0118*	-0.00363
	(0.0111)	(0.0133)	(384.4)	(0.000614)	(0.00650)	(0.0108)
Mean	0.230	0.431	9278.4	0.0161	0.0692	0.169

Sources: CJARS JOE 2023Q3 Commuting Zone Data

Notes: Each point estimate represents a separate difference in differences regression estimated following Gardner (2022) restricted to the population indicated. All regressions include year and commuting zone fixed effects. Gray-shaded columns are subject to additional measurement error due to disclosure pooling rules in CJARS JOE data/ Estimates are population-weighted. Commuting zone clustered standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < .01$

Appendix Table A6: Effects of Ban the Box on One Year Outcomes by Subgroup: Returning Prisoner Sample

	Above poverty	Any W2	W2 Wages	Mortality	New Felony	New Misdemeanor	Reincarceration
Race							
White	0.0129** (0.00517)	0.0301*** (0.00781)	549.5*** (178.3)	0.00253*** (0.000972)	-0.0125 (0.00960)	0.0432** (0.0170)	-0.0204* (0.0105)
Mean	0.206	0.520	7357.4	0.0217	0.144	0.169	0.151
Black	0.00945 (0.00680)	0.0228* (0.0128)	240.4 (207.7)	0.0000383 (0.000679)	-0.0109 (0.0121)	0.0555* (0.0328)	-0.0101 (0.0131)
Mean	0.124	0.457	4260.0	0.0157	0.200	0.217	0.158
Age							
Age 15-24	-0.00104 (0.00893)	0.0322** (0.0126)	-9.382 (316.2)	0.00155 (0.000949)	-0.0121 (0.0123)	0.0656* (0.0354)	-0.0243* (0.0139)
Mean	0.139	0.541	5180.0	0.00839	0.237	0.263	0.184
Age 25-40	0.00736 (0.00712)	0.0319*** (0.00821)	157.5 (244.3)	0.000963 (0.000608)	-0.0145 (0.0115)	0.0490** (0.0222)	-0.0194* (0.0115)
Mean	0.202	0.542	7410.1	0.0106	0.170	0.191	0.156
Age 40+	0.00967* (0.00563)	0.0193** (0.00874)	400.2* (210.9)	0.00239* (0.00128)	-0.0105 (0.00884)	0.0318* (0.0191)	-0.0110 (0.00939)
Mean	0.170	0.409	6024.6	0.0353	0.118	0.132	0.127
Record							
First Time	0.0241*** (0.00528)	0.0340*** (0.00799)	836.5*** (171.6)	0.00138** (0.000637)	-0.0110 (0.00887)	0.0473*** (0.0152)	-0.0362*** (0.00877)
Mean	0.218	0.548	7779.6	0.0164	0.114	0.124	0.110
Repeat	0.0212*** (0.00459)	0.0371*** (0.00837)	757.4*** (192.9)	0.00173* (0.000979)	-0.0121 (0.0116)	0.0646*** (0.0199)	-0.0288*** (0.0111)
Mean	0.148	0.447	5530.6	0.0215	0.195	0.187	0.187

Sources: CJARS JOE 2023Q3 Commuting Zone Data

Notes: Each point estimate represents a separate difference in differences regression estimated following Gardner (2022) restricted to the population indicated. All regressions include year and commuting zone fixed effects. Gray-shaded columns are subject to additional measurement error due to disclosure pooling rules in CJARS JOE data. Estimates are population-weighted. Commuting zone clustered standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < .01$

Appendix Table A7: Effects of Ban the Box on One Year Outcomes by Age and Race: Returning Prisoner Sample

	Above poverty	Any W2	W2 Wages	Mortality	New Felony	New Misdemeanor	Reincarceration
White 15-24	0.00672 (0.00698)	0.0441*** (0.0118)	362.8 (222.1)	0.000704 (0.00149)	-0.00688 (0.0103)	0.0473* (0.0249)	-0.0257* (0.0137)
Mean	0.179	0.616	6248.7	0.00815	0.198	0.236	0.180
Black 15-24	-0.0000738 (0.00607)	0.0149 (0.0207)	-78.95 (220.3)	0.00114 (0.00142)	-0.0170 (0.0185)	0.0708 (0.0477)	-0.0142 (0.0200)
Mean	0.0654	0.447	2672.2	0.00869	0.297	0.311	0.192
White 25-40	0.00904 (0.00615)	0.0316*** (0.00776)	280.7 (210.2)	0.00168 (0.00106)	-0.0166 (0.0111)	0.0476*** (0.0173)	-0.0173 (0.0134)
Mean	0.226	0.571	8038.3	0.0121	0.166	0.191	0.165
Black 25-40	0.0116 (0.00780)	0.0300** (0.0121)	334.7 (235.6)	-0.000327 (0.000676)	-0.00949 (0.0117)	0.0498* (0.0299)	-0.0110 (0.0134)
Mean	0.138	0.498	4692.2	0.00906	0.203	0.222	0.155
White 40+	0.0189*** (0.00511)	0.0235*** (0.00874)	867.6*** (198.5)	0.00397** (0.00170)	-0.0121 (0.00770)	0.0178 (0.0137)	-0.0132* (0.00776)
Mean	0.189	0.414	6831.8	0.0404	0.103	0.120	0.120
Black 40+	0.00996 (0.00746)	0.0173 (0.0109)	320.4 (200.1)	-0.000290 (0.00127)	-0.0121 (0.0113)	0.0446 (0.0276)	-0.00508 (0.0120)
Mean	0.133	0.399	4366.6	0.0286	0.152	0.165	0.143

Sources: CJARS JOE 2023Q3 Commuting Zone Data

Notes: Each point estimate represents a separate difference in differences regression estimated following Gardner (2022) restricted to the population indicated. All regressions include year and commuting zone fixed effects. Gray-shaded columns are subject to additional measurement error due to disclosure pooling rules in CJARS JOE data. Estimates are population-weighted. Commuting zone clustered standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < .01$

Appendix Table A8: Effects of Ban the Box on One Year Outcomes by Treatment-Year Cohorts: Misdemeanor Defendant Sample, Callaway and Sant'Anna (2021)

	Above poverty	Any W2	W2 Wages	Mortality	New Felony	New Misdemeanor
GAverage	0.0163*** (0.00495)	0.0141** (0.00624)	626.7*** (187.8)	-0.000573** (0.000262)	-0.00819*** (0.00189)	-0.00523 (0.00330)
g2007	0.00120 (0.00548)	0.000668 (0.00929)	212.0 (307.2)	-0.00164*** (0.000496)	0.00333* (0.00195)	-0.0154 (0.0140)
g2008	-0.00397 (0.00799)	0.00213 (0.0109)	-295.5** (145.3)	-0.000160 (0.000662)	-0.00679*** (0.00219)	-0.0133*** (0.00278)
g2010	0.0566*** (0.00347)	0.0680*** (0.00547)	1840.2*** (263.3)	-0.000326 (0.000234)	-0.0189*** (0.00289)	-0.0179*** (0.00347)
g2011	0.0177*** (0.00294)	0.0392*** (0.00543)	640.8*** (246.4)	0.00207*** (0.000637)	-0.0175*** (0.00338)	-0.0103*** (0.00380)
g2012	0.0233*** (0.00854)	0.00739 (0.00680)	798.2*** (288.3)	-0.00177*** (0.000217)	-0.00617*** (0.00187)	0.00708 (0.00465)
g2013	0.0130*** (0.00472)	0.0189** (0.00937)	555.4*** (131.0)	-0.000423 (0.000586)	-0.00615*** (0.00206)	0.0118** (0.00547)
g2014	0.0174*** (0.00604)	0.0106*** (0.00356)	886.6** (371.4)	0.0000588 (0.000472)	-0.00973* (0.00527)	-0.0102 (0.00670)
g2015	0.0164** (0.00667)	0.00733 (0.00470)	527.8 (329.1)	-0.000580 (0.000359)	-0.00566 (0.00348)	-0.00768** (0.00361)
g2016	-0.00586 (0.00501)	-0.00735* (0.00432)	-402.6* (239.4)	-0.000117 (0.000728)	-0.00812*** (0.00291)	-0.00125 (0.00529)
g2017	0.0291*** (0.00803)	0.0185*** (0.00386)	1386.8*** (410.8)	-0.00310*** (0.00109)	-0.00488* (0.00270)	0.00812 (0.00527)
g2018	0.00151 (0.00927)	-0.00988*** (0.00374)	484.5 (686.7)	-0.000815 (0.000562)	-0.00463 (0.00339)	-0.0105** (0.00519)

Sources: CJARS JOE 2023Q3 Commuting Zone Data

Notes: Each point estimate represents a separate difference in differences regression estimated following Callaway and Sant'Anna (2021). Estimates are the ATT for each treatment cohort, across all period. All regressions include year and commuting zone fixed effects. Estimates are population-weighted. Commuting zone clustered standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < .01$

Appendix Table A9: Robustness, Leave-one-out Analysis: Effects of Ban the Box on One Year Outcomes (Misdemeanor Defendants)

	State dropped:						
	AZ	FL	MD	MI	MN	NC	ND
Above Poverty	0.0153 (0.00972)	0.0348*** (0.0101)	0.0351*** (0.00998)	0.0280*** (0.00981)	0.0214*** (0.00739)	0.0341*** (0.0102)	0.0282*** (0.00918)
Any W2	0.0186* (0.01000)	0.0364*** (0.0110)	0.0375*** (0.0106)	0.0281*** (0.00969)	0.0239*** (0.00913)	0.0359*** (0.0108)	0.0299*** (0.00977)
W2 Wages	412.3 (422.6)	1316.8*** (504.5)	1311.7** (511.8)	1145.2** (559.7)	659.4*** (227.1)	1269.7** (498.5)	1137.7** (450.0)
Mortality	-0.000680* (0.000357)	-0.000832** (0.000393)	-0.000642* (0.000328)	-0.000874** (0.000400)	-0.000511 (0.000323)	-0.000552 (0.000367)	-0.000851** (0.000340)
Felony Recidivism	-0.00727*** (0.00279)	-0.0138*** (0.00316)	-0.0152*** (0.00266)	-0.0149*** (0.00341)	-0.0116*** (0.00269)	-0.0139*** (0.00319)	-0.0127*** (0.00280)
Misdemeanor Recidivism	0.00447 (0.00653)	0.00622 (0.00851)	0.00722 (0.00763)	0.00817 (0.00863)	0.00106 (0.00708)	-0.00802 (0.00525)	0.00809 (0.00703)

	State dropped:					
	OK	OR	PA	TX	VA	WI
Above Poverty	0.0308*** (0.00925)	0.0318*** (0.00945)	0.0313*** (0.00948)	0.0316*** (0.00941)	0.0341*** (0.0105)	0.0315*** (0.00950)
Any W2	0.0330*** (0.00984)	0.0345*** (0.00994)	0.0333*** (0.0101)	0.0343*** (0.00998)	0.0377*** (0.0109)	0.0340*** (0.0101)
W2 Wages	1182.9*** (453.2)	1230.7*** (463.4)	1194.6** (464.3)	1350.9*** (452.5)	1346.5*** (516.2)	1215.7*** (464.2)
Mortality	-0.000733** (0.000343)	-0.000757** (0.000349)	-0.000773** (0.000340)	-0.000965*** (0.000329)	-0.000608 (0.000387)	-0.000838** (0.000347)
Felony Recidivism	-0.0133*** (0.00284)	-0.0139*** (0.00285)	-0.0136*** (0.00300)	-0.0135*** (0.00287)	-0.0152*** (0.00306)	-0.0136*** (0.00284)
Misdemeanor Recidivism	0.00507 (0.00724)	0.00459 (0.00736)	0.00623 (0.00771)	0.00507 (0.00721)	0.00546 (0.00821)	0.00510 (0.00730)

Sources: CJARS JOE 2023Q3 Commuting Zone Data

Notes: Each point estimate represents a separate difference in differences regression estimated following Gardner (2022) restricted to the population dropping the state indicated. Estimates are population-weighted. All regressions include year and commuting zone fixed effects. Estimates are population-weighted. Commuting zone clustered standard errors in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < .01$

Appendix Table A10: Effects of Ban the Box on One Year Outcomes With Controls

Panel A: Misdemeanor Defendant Sample

	Above poverty	Any W2	W2 Wages	Mortality	New Felony	New Misdemeanor
Ban the Box	0.0377*** (0.0110)	0.0447*** (0.0118)	1645.4*** (598.4)	-0.00127*** (0.000454)	-0.0167*** (0.00600)	0.000466 (0.00898)
Mean	0.312	0.614	12719.9	0.00941	0.0735	0.169

Panel B: Felony Defendant Sample

	Above Poverty	Any W2	W2 Wages	Mortality	New Felony	New Misdemeanor
Ban the Box	0.0103*** (0.00330)	0.0220*** (0.00614)	234.7* (124.3)	0.00000806 (0.000357)	-0.00410 (0.00410)	0.0104 (0.0125)
Mean	0.164	0.452	6468.5	0.00983	0.115	0.175

Panel C: Returning Prisoner Sample

	Above poverty	Any W2	W2 Wages	Mortality	New Felony	New Misdemeanor	Reincarceration
Ban the Box	0.00135 (0.00665)	0.0166* (0.00967)	183.2 (281.1)	0.000956 (0.000730)	-0.0215** (0.00940)	0.122*** (0.0464)	-0.00528 (0.0158)
Mean	0.183	0.500	6645.9	0.0191	0.159	0.178	0.149

Sources: CJARS JOE 2023Q3 Commuting Zone Data; LEHD QWI; BLS LAUS; NIH SEER Population Estimates
Notes: Each point estimate represents a separate difference in differences regression estimated following Gardner (2022) restricted to the population indicated. All regressions include year and commuting zone fixed effects. These tables additionally control for the commuting zone's unemployment rate, labor force participation, the log of population, and the hiring ratios included in Table 1. Estimates are population-weighted. Commuting zone clustered standard errors in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < .01$

Appendix Table A11: Other outcomes around BTB adoption (imputation)

	Misdemeanor Defendants			Felony Defendants			Returning Prisoners	
	Cohort Size	Rate	Convicted	Cohort Size	Rate	Convicted	Cohort Size	Rate
BTB	-7759.8 (4901.7)	-33.22 (177.1)	0.0320 (0.0254)	-2468.0 (1908.0)	-235.1*** (74.12)	0.0516** (0.0202)	-493.3 (395.7)	-22.27 (30.55)

Sources: CJARS JOE 2023Q3 Commuting Zone Data

Notes: Each point estimate represents a separate difference in differences regression estimated following Gardner (2022) restricted to the population indicated. All regressions include year and commuting zone fixed effects. Estimates are population-weighted. Commuting zone clustered standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < .01$

Appendix Table A12: Other outcomes around BTB adoption (imputation)

	Emp. Rate	LFP	HS Hired	HS Const. Hired	Col Hired	ln(HS Wage)	ln(Col. Wage)
BTB	-0.00109 (0.00183)	0.00858*** (0.00223)	0.000869** (0.000379)	0.000188 (0.000168)	0.000514 (0.000312)	0.00369 (0.0111)	0.0193 (0.0201)

Sources: CJARS JOE 2023Q3 Commuting Zone Data; LEHD QWI; BLS LAUS; NIH SEER Population Estimates

Notes: Each point estimate represents a separate difference in differences regression estimated following Gardner (2022) restricted to the population indicated. All regressions include year and commuting zone fixed effects. Estimates are population-weighted. Commuting zone clustered standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < .01$

Appendix Table A13: Effects of Ban the Box on One Year Outcomes: Public and Private Laws

Panel A: Misdemeanor Defendant Sample

	Above Poverty	Any W2	W2 Wages	Mortality	New Felony	New Misdemeanor
BTB (private)	0.0399 (0.0295)	0.0449* (0.0231)	2607.9 (1717.5)	-0.000830 (0.00130)	-0.0101* (0.00546)	-0.00147 (0.0166)
Ban the Box	0.0272*** (0.00807)	0.0290*** (0.00960)	947.8*** (317.6)	-0.000677** (0.000330)	-0.0135*** (0.00264)	0.00656 (0.00759)
Mean	0.312	0.614	12719.9	0.00941	0.0735	0.169

Panel B: Felony Defendant Sample

	Above Poverty	Any W2	W2 Wages	Mortality	New Felony	New Misdemeanor
BTB (private)	-0.0104 (0.00734)	-0.00297 (0.0113)	-322.0 (240.4)	0.000523 (0.00148)	-0.00610 (0.0103)	-0.00943 (0.0270)
Ban the Box	0.00886** (0.00435)	0.0112 (0.00817)	222.5 (136.3)	-0.000242 (0.000464)	-0.0101** (0.00475)	0.0121 (0.0128)
Mean	0.164	0.452	6468.5	0.00983	0.115	0.175

Panel C: Returning Prisoner Sample

	Above Poverty	Any W2	W2 Wages	Mortality	New Felony	New Misdemeanor	Reincarceration
BTB (private)	-0.0417** (0.0170)	-0.0111 (0.0221)	-1395.4** (601.1)	-0.00165 (0.00104)	0.0234 (0.0268)	0.0118 (0.0509)	0.0241 (0.0224)
Ban the Box	0.0169*** (0.00505)	0.0253*** (0.00746)	599.7*** (184.2)	0.00194*** (0.000703)	-0.0175* (0.0100)	0.0451* (0.0250)	-0.0248** (0.00977)
Mean	0.183	0.500	6645.9	0.0191	0.159	0.178	0.149

Sources: CJARS JOE 2023Q3 Commuting Zone Data

Notes: Each column represents a separate difference in differences regression estimated following Gardner (2022). All regressions include year and commuting zone fixed effects. Estimates are population-weighted. Commuting zone clustered standard errors in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < .01$